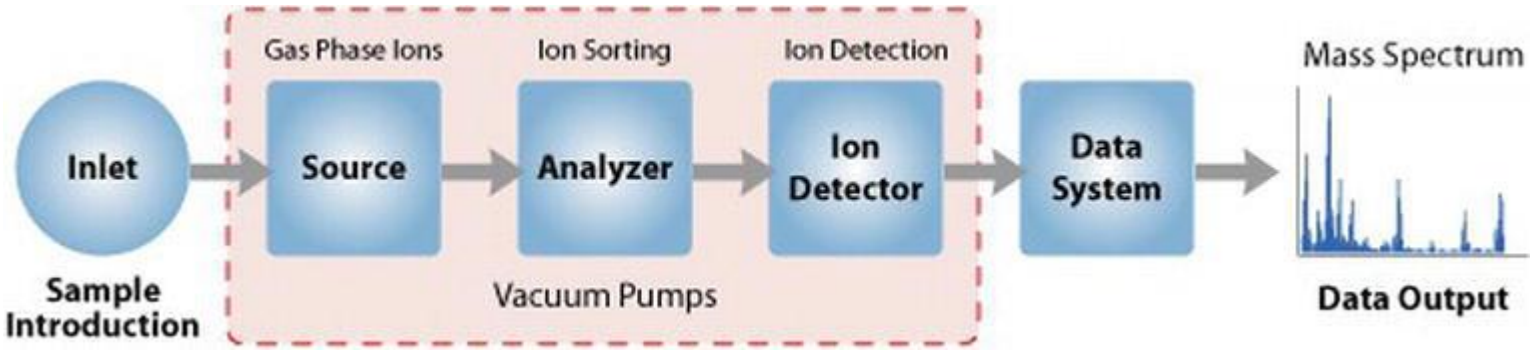
의약품분석학II
- Lecture 5 -

의약품기기분석학

제1장. 분광분석법

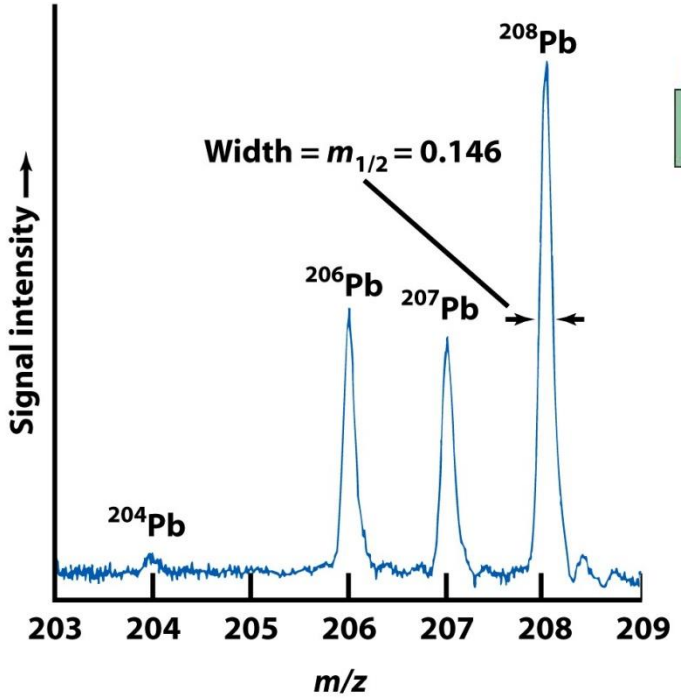
1.9 질량분석법



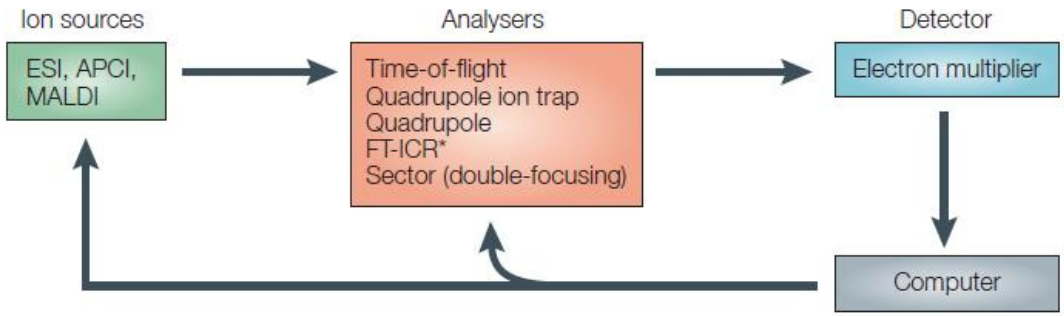
What is mass spectrometry

A technique for studying the masses of atoms, molecules or fragments of molecules

Gas-phase analytes are ionized → **Ions are accelerated by an electric field** → **Ions are separated according to their mass-to-charge ratio (m/z)** → **Mass spectrum**



Mass spectrum showing natural isotopes of Pb



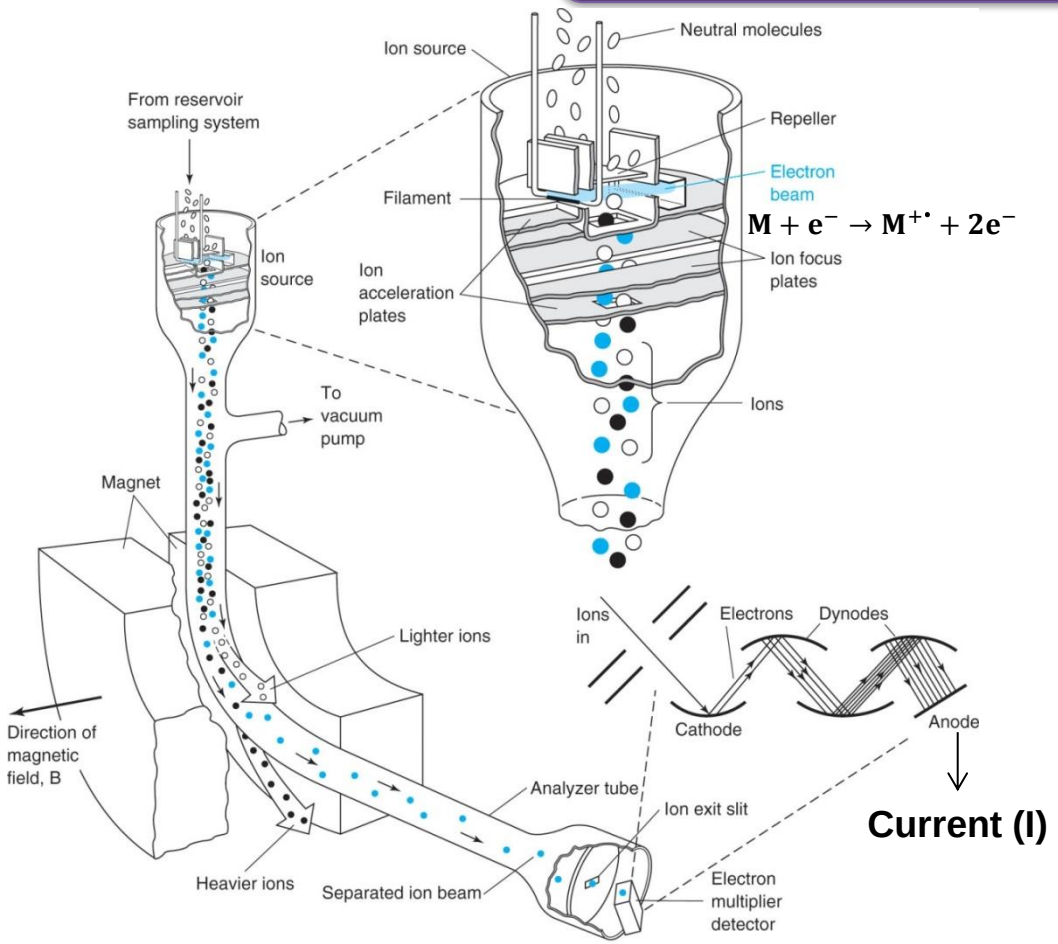
MW 500	MW 1000
M^+ : *m/z* 500	*m/z* 1000
M^{2+} : *m/z* 250	*m/z* 500
M^- : *m/z* 500	*m/z* 1000
M^{2-} : *m/z* 250	*m/z* 500

Unit of mass-to-charge ratio (*m/z*): Thomson (Th)

What is mass spectrometry

이온화부 | 검출기

Ion source | **Mass analyzer (질량분석관)** | **Detector**

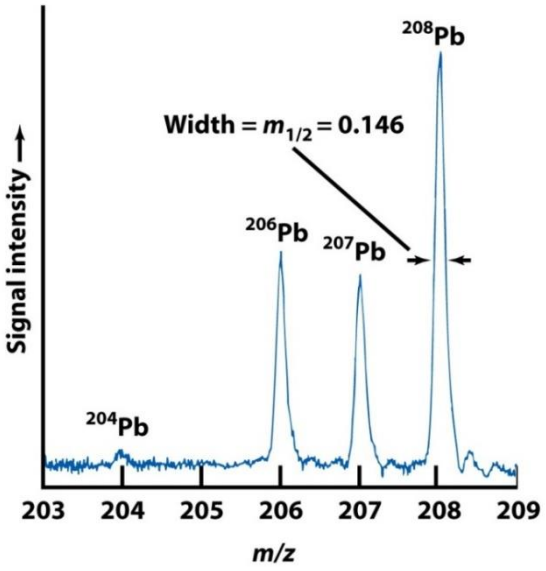


1. **Gaseous molecules are ionized by electron beam, i.e. electron ionization**
2. **Ions are accelerated by an electric field, and expelled into the mass analyzer tube (with same kinetic energy)**
3. **Accelerated ions are deflected in the magnetic field perpendicular to their direction of travel**
4. **Mass spectrum is obtained by varying the magnetic field strength: heavy ions are not deflected enough and light ions are deflected too much to reach the detector**

Magnetic sector mass spectrometer(직교자기장형 질량분석기)
(ions with different mass, but constant kinetic energy, are separated by their trajectories in a magnetic field)

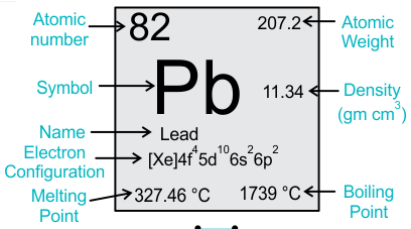
What is mass spectrometry

Molecular mass and nominal mass



Atomic mass (weight): weighted average of natural isotopes of an element

^{204}Pb: 203.97303 amu, u (or Da) (1.4%)
^{206}Pb: 205.97445 u (24.1%)
^{207}Pb: 206.97588 u (22.1%)
^{208}Pb: 207.97664 u (52.4%)



Atomic mass of Pb = 203.97303 x 0.014
+ 205.97445 x 0.241 + 206.97558 x 0.221 + 207.97664 x 0.524
= 207.21683

^{79}Br: 78.91834 u (50.69%)
^{81}Br: 80.91629 u (49.31%)
Atomic mass of Br =78.91834x0.5069+80.91629x0.4931=79.904

Molecular mass (weight): sum of atomic masses listed in the periodic table

Molecular mass of C_2H_5Br = (2x12.0107)+(5x1.00794)+(1x79.904)
= 108.965

1 amu (atomic mass unit), u or Da
= 1/12 of mass of ^{12}C
= 1.66 x 10^{-27} kg
Mass of ^{12}C = 12.0000 u
Mass of ^{1}H = 1.007825 u
Mass of 1 mole of ^{12}C
= 12.0000 u x (1.66x10^{-27} kg/u) x (6.022x10^{23}/mol)
= 0.0120 kg/mol = 12.0 g/mol

Nominal mass(명목질량): sum of the mass numbers(양성자+중성자) of the most abundant isotope of each constituent atoms of a molecule

Nominal mass of C_2H_5Br = (2x12)+(5x1)+(1x79)
= 108

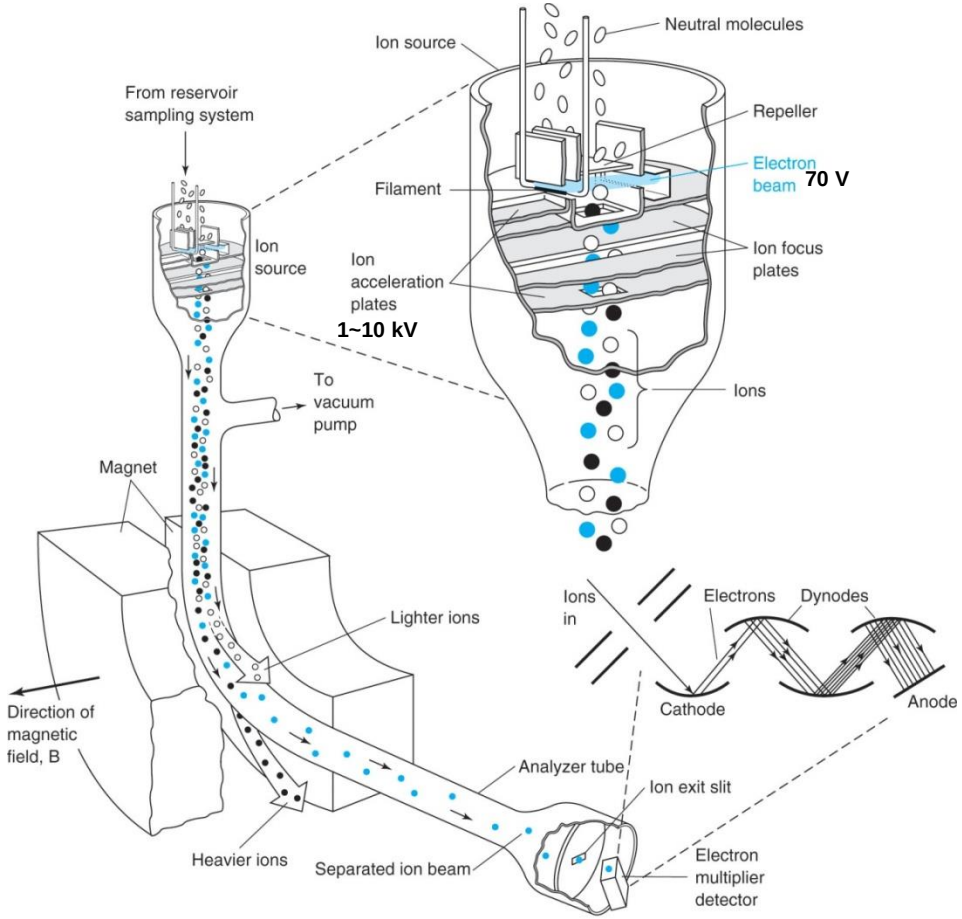
Monoisotopic mass(단일 동위원소 질량): sum of masses of the most abundant natural isotope of each constituent atoms of a molecule

Monoisotopic mass of C_2H_5Br = (2x12.0000)+(5x1.00782)+(1x 78.91834)
= 107.95751

What is mass spectrometry

Electron ionization (EI) 전자이온화법





Magnetic sector mass spectrometer

1. **Electrons emitted from a hot filament are accelerated through 70 V gaining kinetic energy (E_k) for 70 eV**

$W(J) = E(V) \cdot q(C)$: work done when a charge, q, moves through a potential difference, E

$E_k = eV$

e: magnitude of charge of an electron ($1.602176634 \times 10^{-19}$ C)

V: electric potential (voltage) difference

2. **Some (~0.01%) molecules (M) absorb as much as 12-15 eV when collide with electron beam, which is enough for ionization**

3. **Generated molecular ion ($M^{+\cdot}$) usually has enough internal energy (~1 eV) to break into fragments**

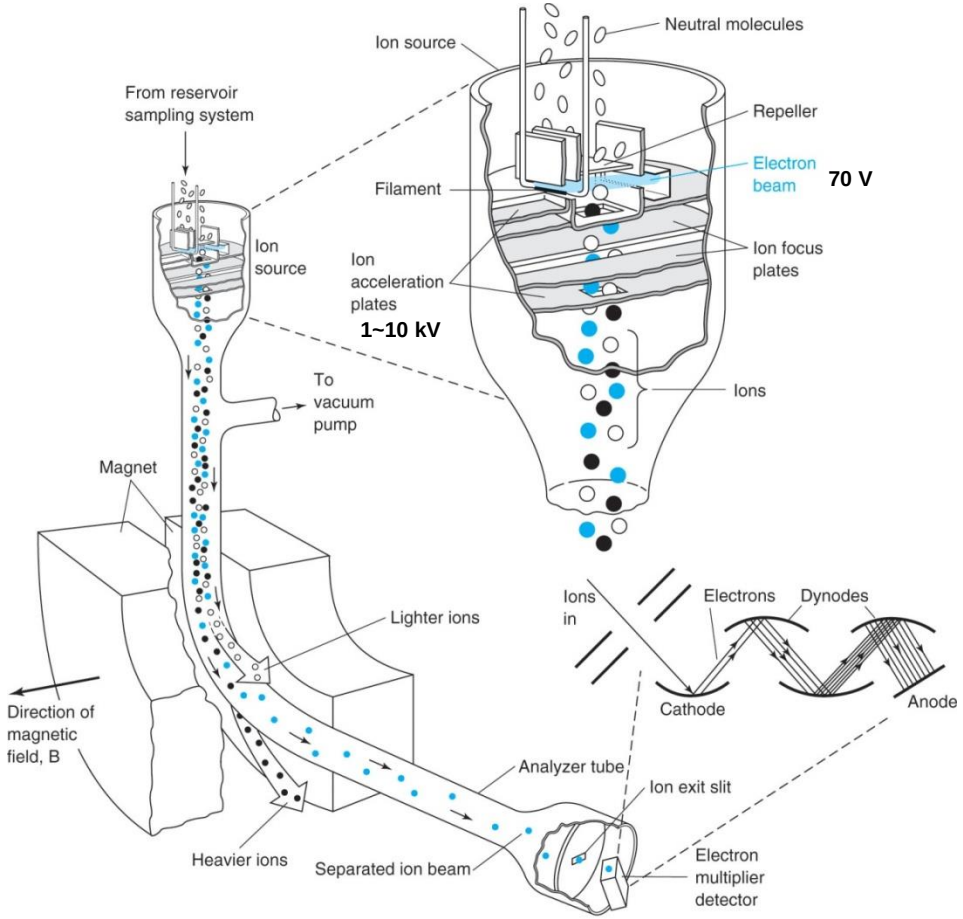
$$M + e^- \rightarrow M^{+\cdot} + e^- + e^-$$

70 eV — molecular ion — ~55 eV — 0.1 eV

What is mass spectrometry

Electron ionization (EI) 전자이온화법



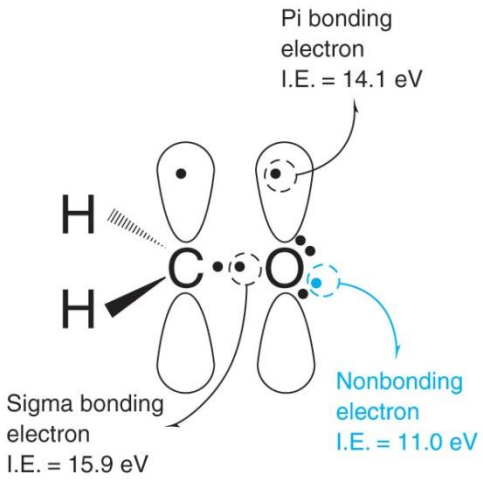


Magnetic sector mass spectrometer

4. **A small positive potential on the repeller plate pushes ions toward the analyzer tube**
5. **A small potential on the ion focus plates creates a focused ion beam**
6. **High voltage (~1-10 kV) between the ion acceleration plates imparts a high velocity to ions as they are expelled from the ion source**

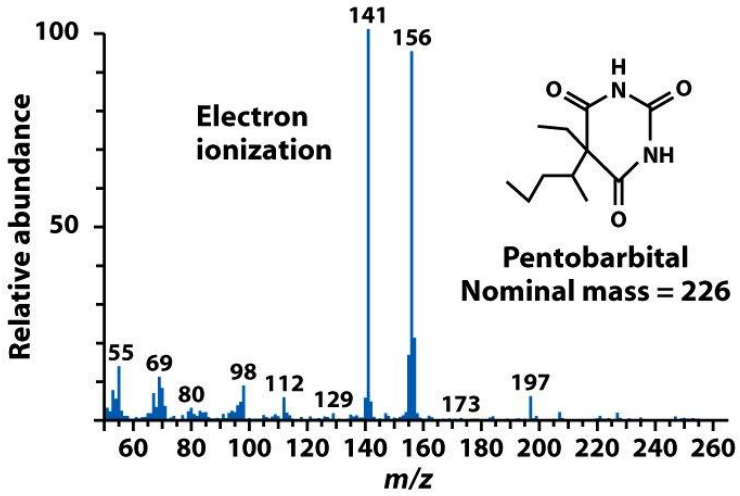
What is mass spectrometry

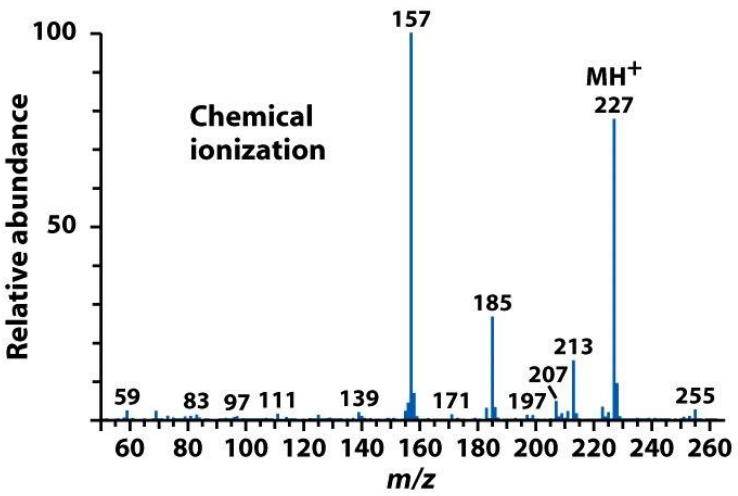
Electron ionization (EI) 전자이온화법



I.E. : ionization energy

- Interaction of a molecule with a 70-eV electron most likely removes the electron with lowest ionization energy
- The resulting molecular ion ($M^{+\bullet}$) can have so much extra energy that it breaks into fragments

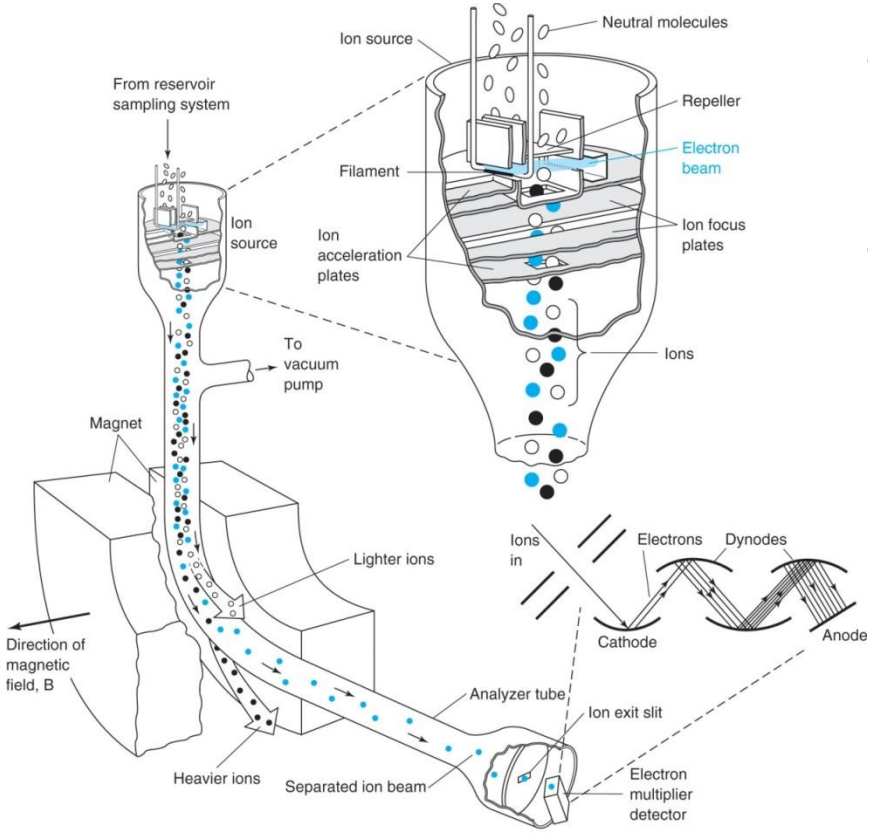




Base peak: the most intense peak in a mass spectrum
Intensities of other peaks are expressed as a percentage of the base peak

What is mass spectrometry

Chemical ionization (CI) 화학이온화법



Ion source is filled with a reagent gas (e.g. methane, isobutane or ammonia) at ~1 mbar

Energetic electrons (100~200 eV) convert reagent gas molecule into various reactive products (reagent ions)

$$CH_4 + e^- \rightarrow CH_4^{+\cdot} + 2e^-$$

$$CH_4^{+\cdot} + CH_4 \rightarrow CH_5^+ + {}^{\cdot}CH_3$$

$$CH_4^{+\cdot} \rightarrow CH_3^+ + {}^{\cdot}H$$

$$CH_3^+ + CH_4 \rightarrow C_2H_5^+ + H_2$$

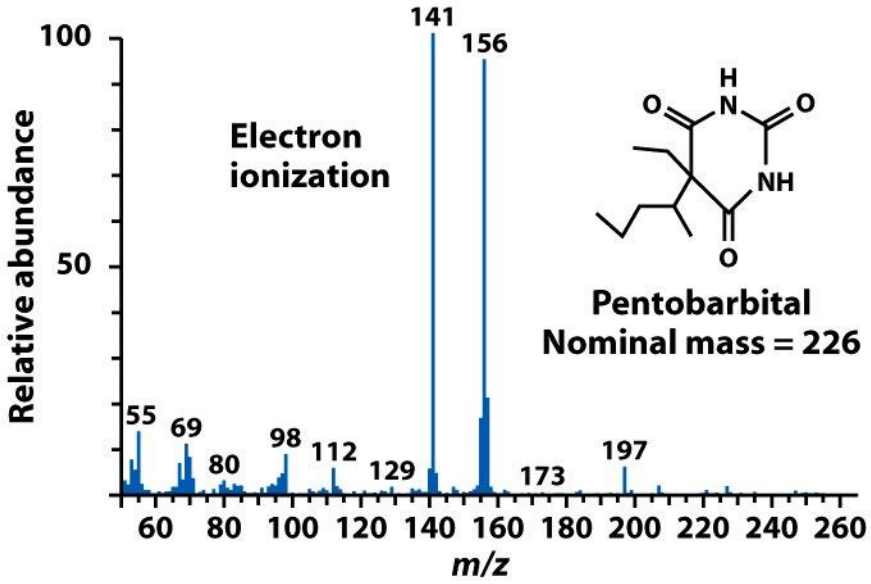
$$CH_5^+ + M \rightarrow CH_4 + MH^+$$

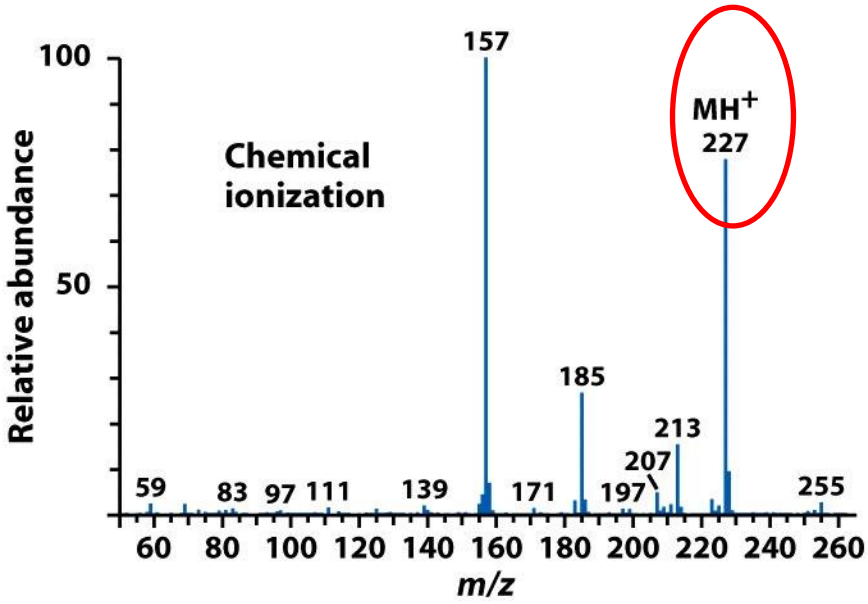
Quasi-molecular ion(준분자 이온)

Quasi-molecular ion: protonated ($[M+H]^+$ and deprotonated ($[M-H]^-$ molecules)

What is mass spectrometry

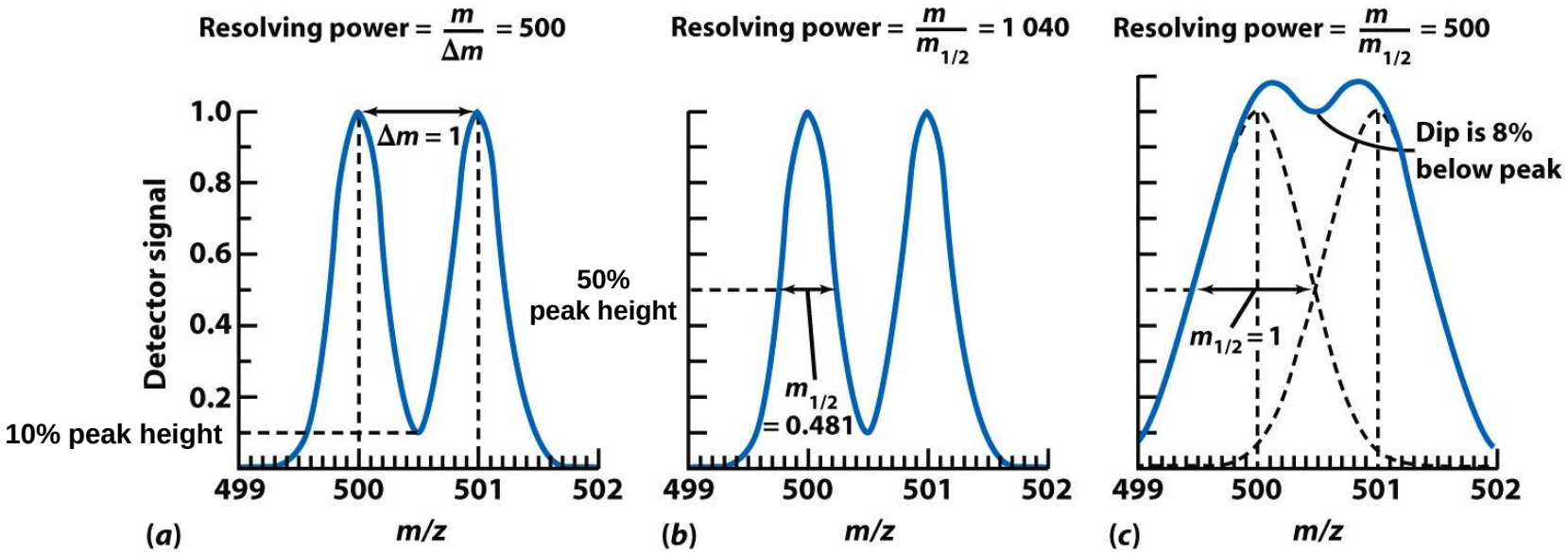
Chemical ionization (CI) 화학이온화법





What is mass spectrometry

Resolving power(분해능)



$$\text{Resolving power (RP)} = \frac{\text{m}}{\Delta\text{m}} \text{ or } \frac{\text{m}}{\text{m}_{1/2}}$$

$$\text{Resolution } (\Delta\text{m}) = \frac{\text{m}}{\text{Resolving power}}$$

$$\textbf{Resolution (R)} = \frac{\textbf{m}}{\textbf{FWHM}}$$

m: the smaller value of m/z

Δm: separation of the two peaks when the overlap at their base is 10% of the maximum peak height (a.k.a. 10% valley)

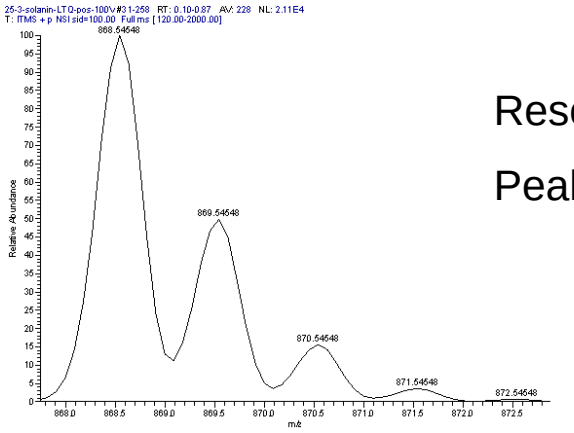
$m_{1/2}$: full width at half maximum (FWHM)

Unit mass resolution: Δm=1 (transmission quadrupole and ion trap mass spectrometers)

Resolution: the smallest mass difference in m/z that can be detected as separate peaks and should be reported with the m/z value, e.g. Δm = 0.1 at m/z = 500

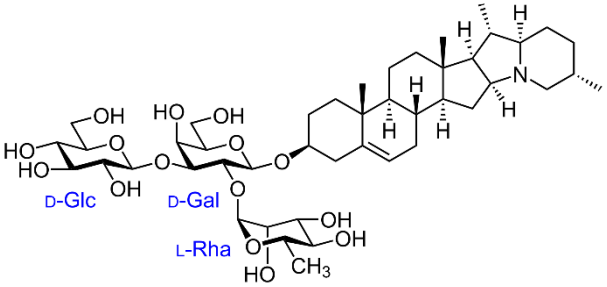
If we have resolving power (RP) ($\frac{m}{\Delta m}$) 10,000, we can separate:

- m/z 50.000 from 50.005 (R = m/RP = 50/10,000 = 0.005)
- m/z 100.000 from 100.010 (R = m/RP = 100/10,000 = 0.01)
- m/z 1,000.000 from 1000.100 (R = m/RP = 1,000/10,000 = 0.1)

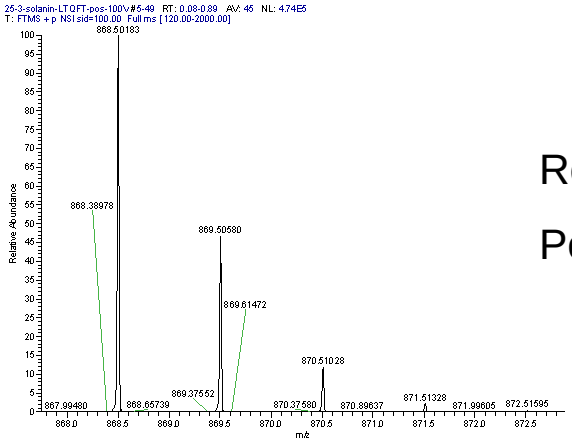


Resolving power ($\frac{m}{m_{1/2}}$) = 1737 at m/z 868.5

Peak width = 0.5 (FWHM)



α-Solanine



Resolving power ($\frac{m}{m_{1/2}}$) = 48,250 at m/z 868.5

Peak width = 0.018 (FWHM)

Isotoplogues: chemical species that differ only in the isotopic composition of their molecules or ions

Interpretation of mass spectrum

The nitrogen rule

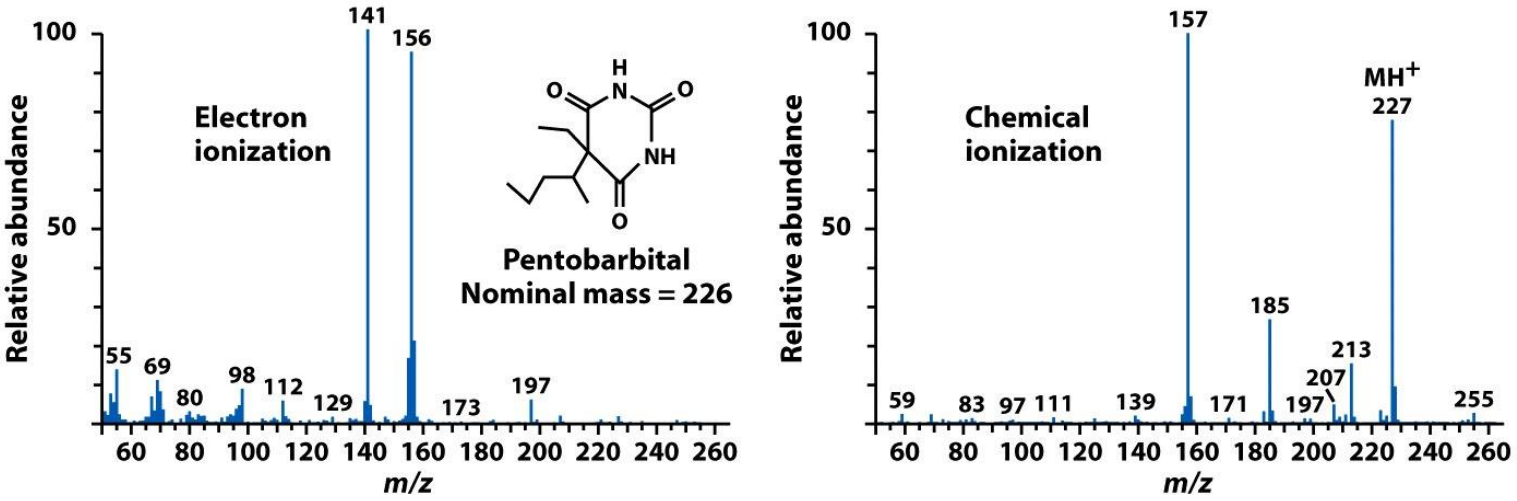
	Mass number	Valency
^{12}C	12	4
^{1}H	1	1
^{14}N	14	3
^{16}O	16	2
^{32}S	32	2
^{19}F	19	1
^{35}Cl	35	1
^{79}Br	79	1
^{127}I	127	1
^{31}P	31	3, 5
^{28}Si	28	4
^{80}Se	80	2

For molecular ion ($M^{+\bullet}$; odd electron ion),

- **Odd number of nitrogen atoms → odd nominal mass**
- **Even number of nitrogen atoms → even nominal mass**

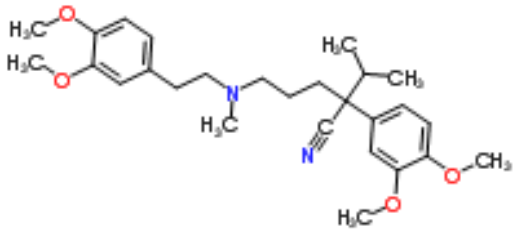
For quasi-molecular ions ($[M+H]^+$, $[M-H]^-$; even electron ions),

- **Odd number of nitrogen atoms → even nominal mass**
- **Even number of nitrogen atoms → odd nominal mass**



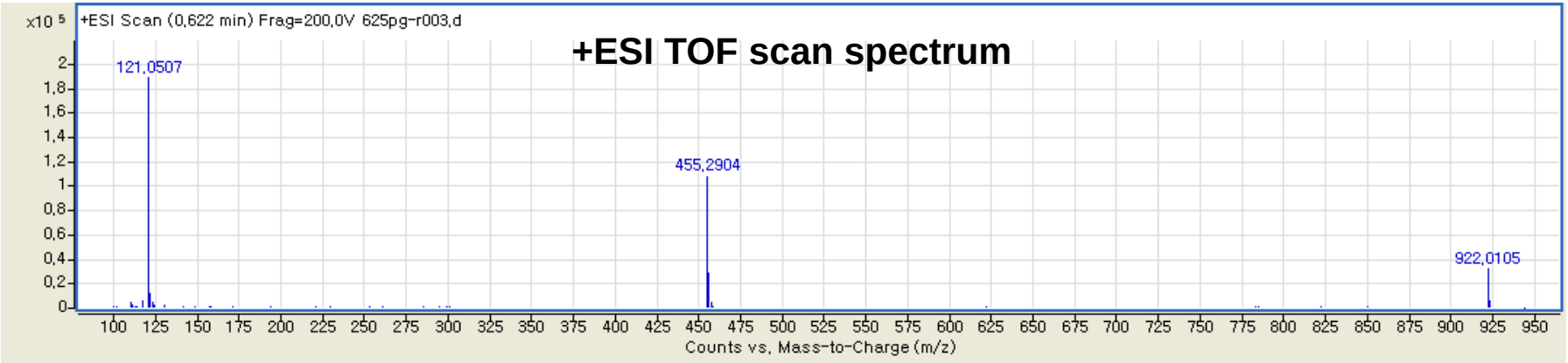
Interpretation of mass spectrum

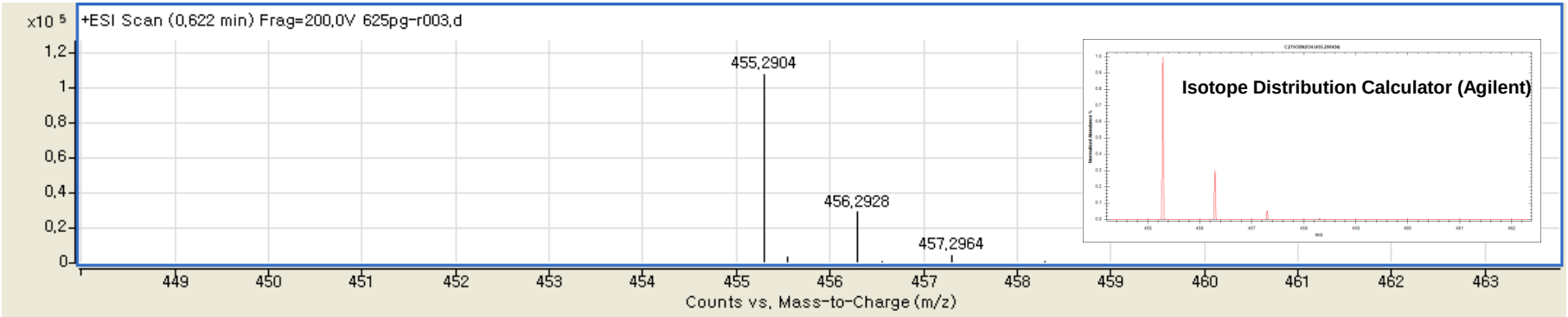
Molecular ion and isotope patterns



Verapamil

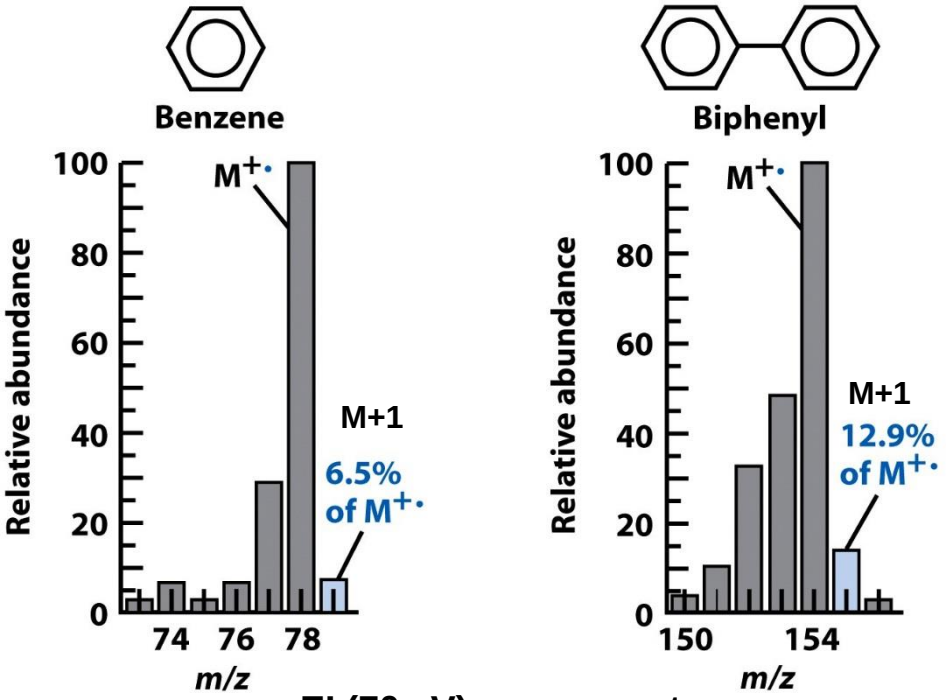
- **Mol. formula: $C_{27}H_{38}N_2O_4$**
- **Monoisotopic mass: 454.2832**
- **Nominal mass: 454**
- **$[M+H]^+$ *m/z* 455.2904**





Interpretation of mass spectrum

Molecular ion and isotope patterns



EI (70 eV) mass spectra

Q: What are nominal masses of the molecular ions of benzene and biphenyl?

Q: Do these results obey the nitrogen rule?

Intensity of M+1 relative to molecular ion for C_nH_m:
Intensity = n x 1.08% + m x 0.012%

For benzene (C_6H_6): Intensity = 6x1.08%+6x0.012% = 6.55%
For biphenyl ($C_{12}H_{10}$): Intensity = 12x1.08%+10x0.012% = 13.08%

$$\text{No. of C} \approx \frac{(\text{M}+1)\ \text{intensity}\ (\%\ \text{of M})}{1.08\%}$$

Interpretation of mass spectrum

Molecular ion and isotope pattern

TABLE 21-1 Isotopes of selected elements

Element	Mass number	Mass (Da)[a]	Abundance (atom%)[b]	Element	Mass number	Mass (Da)[a]	Abundance (atom%)[b]
Proton	1	1.007 276 467	—	Cl	35	34.968 85	75.78
Neutron	1	1.008 664 916	—		37	36.965 90	24.22
Electron	—	0.000 548 580	—	Ar	36	35.967 55	0.336
H	1	1.007 825	99.988		38	37.962 73	0.063
	2	2.014 10	0.012		40	39.962 38	99.600
B	10	10.012 94	19.9	Fe	54	53.939 61	5.845
	11	11.009 31	80.1		56	55.934 94	91.754
C	12	12(exact)	98.93		57	56.935 40	2.119
	13	13.003 35	1.07		58	57.933 28	0.282
N	14	14.003 07	99.632	Br	79	78.918 34	50.69
	15	15.000 11	0.368		81	80.916 29	49.31
O	16	15.994 91	99.757	I	127	126.904 47	100
	17	16.999 13	0.038	Hg	196	195.965 81	0.15
	18	17.999 16	0.205		198	197.966 75	9.97
F	19	18.998 40	100		199	198.968 26	16.87
Si	28	27.976 93	92.230		200	199.968 31	23.10
	29	28.976 49	4.683		201	200.970 29	13.18
	30	29.973 77	3.087		202	201.970 63	29.86
P	31	30.973 76	100		204	203.973 48	6.87
S	32	31.972 07	94.93	Pb	204	203.973 03	1.4
	33	32.971 46	0.76		206	205.974 45	24.1
	34	33.967 87	4.29		207	206.975 88	22.1
	36	35.967 08	0.02		208	207.976 64	52.4

a. 1 dalton (Da) ≡ 1/12 of the mass of ^{12}C = 1.660 538 782 (83) × 10^{-27} kg (from http://physics.nist.gov/constants). Nuclide masses from G. Audi, A. H. Wapsta, and C. Thibault, Nucl. Phys. ***2003,*** *A729, 337, found at www.nndc.bnl.gov/masses. This source provides more significant figures for atomic mass than are cited in this table.*

b. Abundance is representative of what is found in nature. Significant variations are observed. For example, ^{18}O in natural substances has been found in the range 0.188 to 0.222 atom%. The latest list of isotope abundances, which is slightly different from this table, is found in J. K. Böhlke et al., "Isotopic Compositions of the Elements, 2001," J. Phys. Chem. Ref. Data ***2005,*** *34, 57.*

Interpretation of mass spectrum

Molecular ion and isotope pattern

TABLE 21-2 Isotope abundance factors (%) for interpreting mass spectra

Element	X+1	X+2	X+3	X+4	X+5	X+6
H	$0.012n$					
C	$1.08n$	$0.005\ 8n(n-1)$				
N	$0.369n$					
O	$0.038n$	$0.205n$				
F	0					
Si	$5.08n$	$3.35n$	$0.170n(n-1)$	$0.056n(n-1)$		
P	0					
S	$0.801n$	$4.52n$	$0.036n(n-1)$	$0.102n(n-1)$		
Cl	—	$32.0n$	—	$5.11n(n-1)$	—	$0.544n(n-1)(n-2)$
Br	—	$97.3n$	—	$47.3n(n-1)$	—	$15.3n(n-1)(n-2)$
I	0					

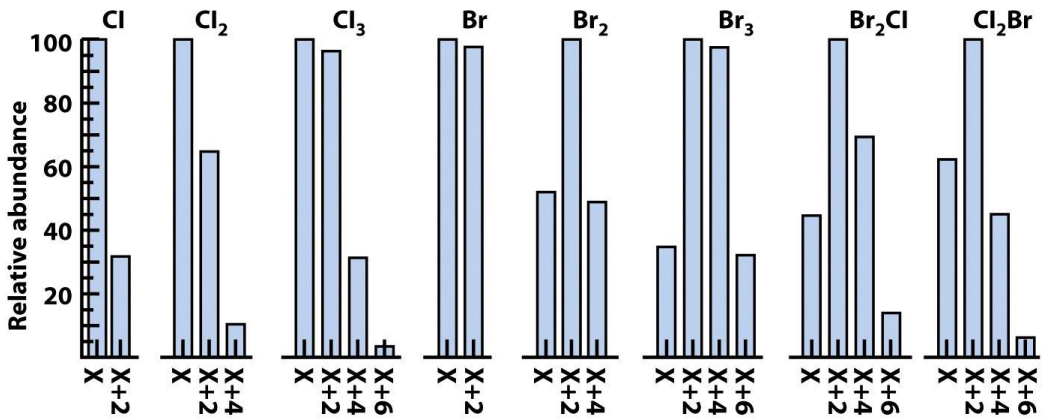
EXAMPLE: For a peak at m/z = X containing n carbon atoms, the intensity from carbon at X+1 is n × 1.08% of the intensity at X. The intensity at X + 2 is n(n − 1) × 0.005 8% of the intensity at X. Contributions from isotopes of other atoms in the ion are additive.

Intensity of M+2 relative to molecular ion (M) for C_6H_6
= 0.0058%x6x5 = 0.17%

Intensity of M+2 relative to molecular ion (M) for Br_2Cl
= 97.3%x2 + 32%x1 = 226.6%

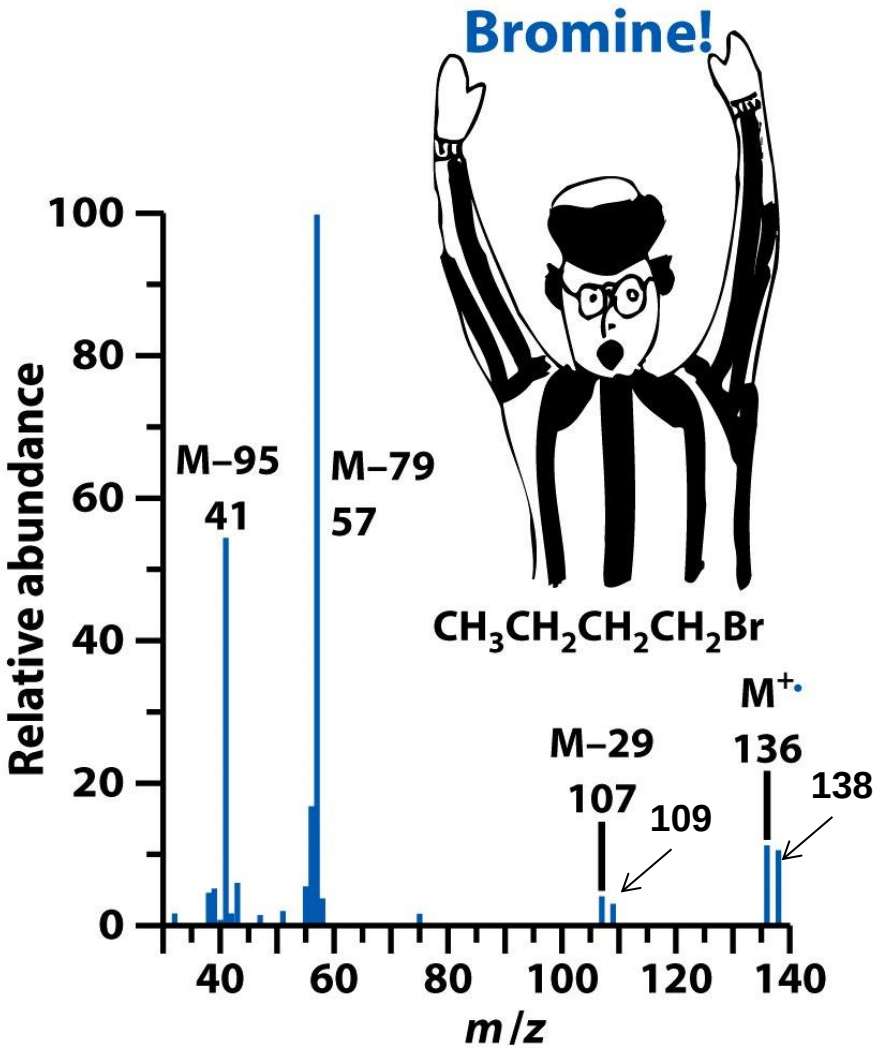
Intensity of M+4 relative to molecular ion for Br_2Cl
= P [($^{81}Br \cap {}^{81}Br$) U ((^{81}Br U ^{81}Br) ∩ ^{37}Cl)]
= 47.3%x2x1 + (0.973x2)x0.32x100% = 156.9%

Intensity of M+6 relative to molecular ion for Br_2Cl
= P ($^{81}Br \cap {}^{81}Br \cap {}^{37}Cl$)
= 0.973x0.973x0.32x100% = 30.3%



Interpretation of mass spectrum

Molecular ion and isotope pattern



Intensity of M+2 (m/z 138)
= (0.0058x4x3)+(97.3x1) = 97.4

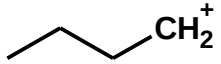
Intensity of M+1 (m/z 137)
= (1.08x4)+(0.012x9) = 4.4%

$$\mathbf{M} \rightarrow \mathbf{M}^{+\cdot} + \mathbf{e}^{-}$$
$$\mathbf{M}^{+\cdot} \rightarrow \mathbf{F}^{+} + \mathbf{N}^{\cdot}$$

M·+ (m/z 136) 

M-29 (m/z 107) 

$+ \mathrm{CH_3CH_2^{\cdot}}$

M-79 (m/z 57) 

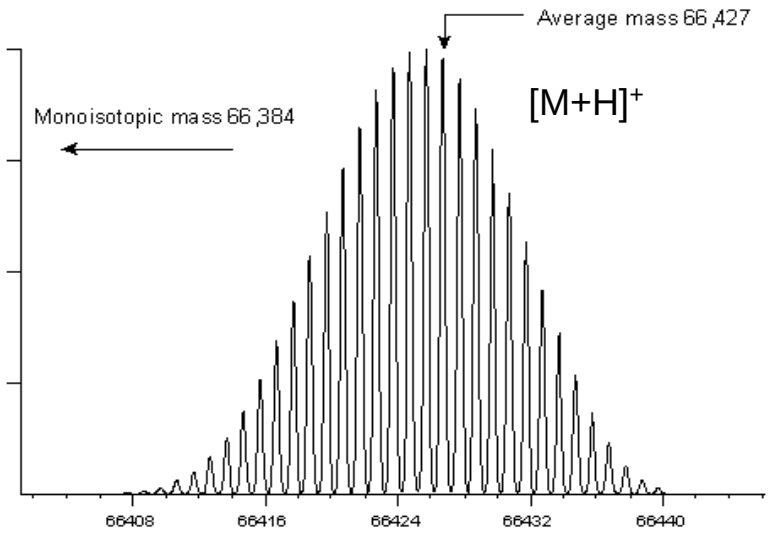
$+ \mathrm{Br}^{\cdot}$

M-95 (m/z 41) 

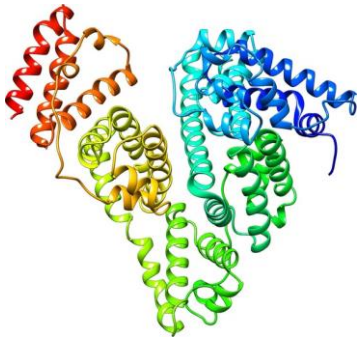
$+ \mathrm{CH_4} + \mathrm{Br}^{\cdot}$

Isotopic envelopes

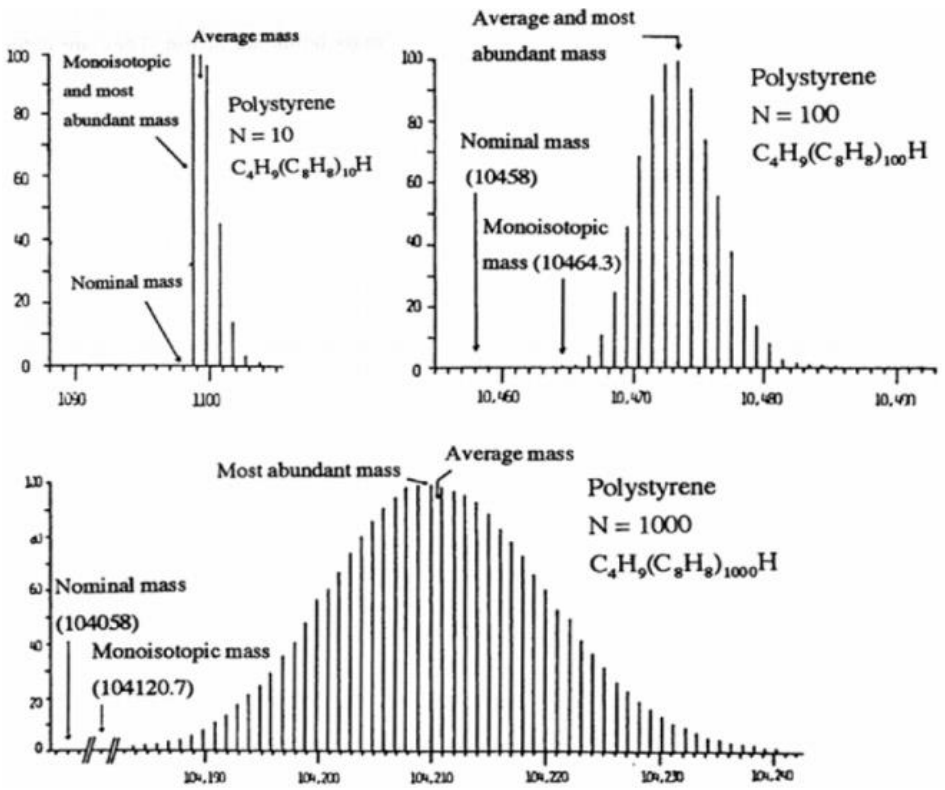
Bovine serum albumin (BSA)

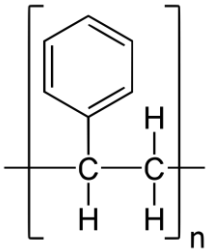


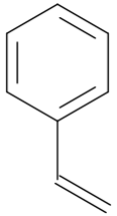
- Number of amino acid residues: 583
- Molecular mass: 66.5 kDa
- Molecular formula: $\sim C_{2934}H_{4581}N_{781}O_{897}S_{39}$



Polystyrene (PS)







Interpretation of mass spectrum

High-resolution mass spectrometry

$C_5H_8O^{+\cdot}$

Nominal mass: 5x12+1x8+1x16=84 Da

Exact mass (monoisotopic mass):

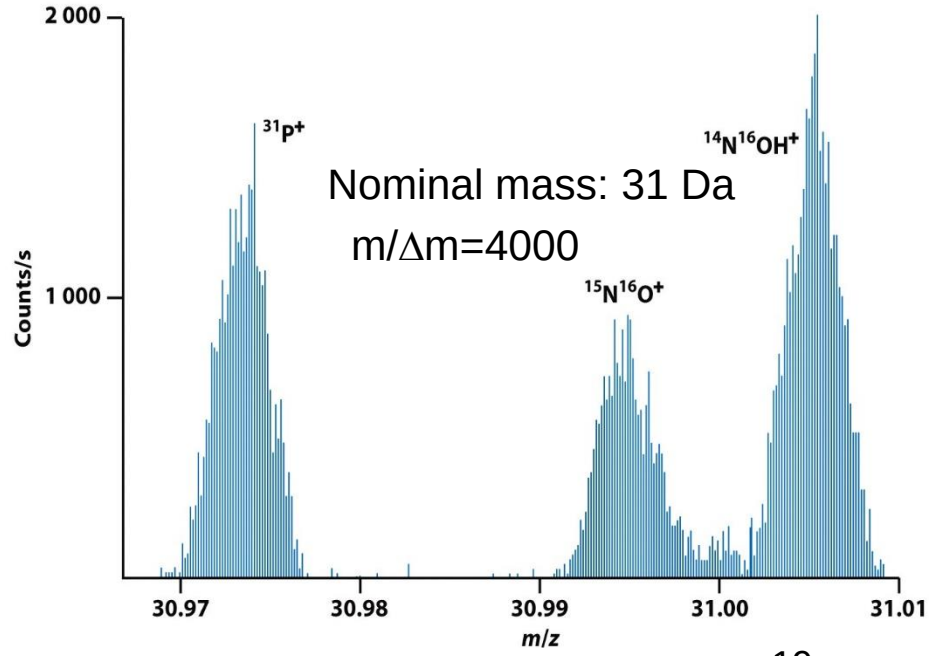
5x12.00000+8x1.007825+1x15.99491-e^- (0.00055)=84.05696 Da

$C_6H_{12}^{+\cdot}$

Nominal mass: 6x12+12x1=84 Da

Exact mass (monoisotopic mass):

6x12.00000+12x1.007825-e^- (0.00055)= 84.09335 Da

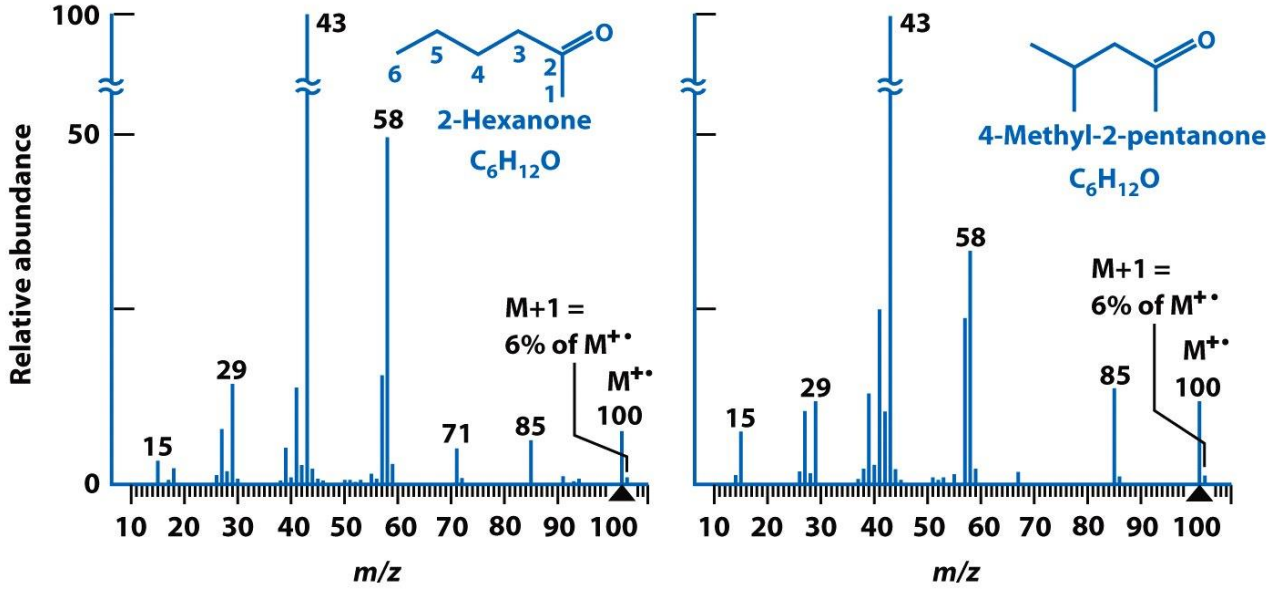


High-resolution double-focusing magnetic sector mass spectrum

m/Δm = 4000

Interpretation of mass spectrum

Identifying the molecular ion peak



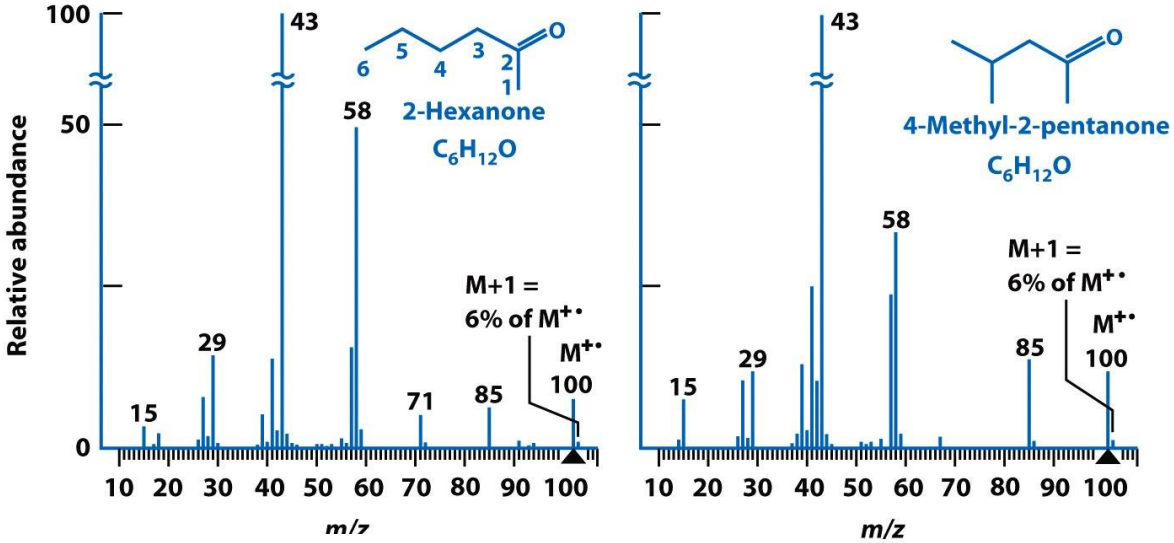
1. M^+ will be at the highest *m/z* value of any of the significant peaks in the spectrum
2. Isotopic patterns must be consistent with the proposed formula
3. The peak for the heaviest fragment ion should not correspond to an improbable mass loss from M^+ (e.g. 3~14, 21~25)

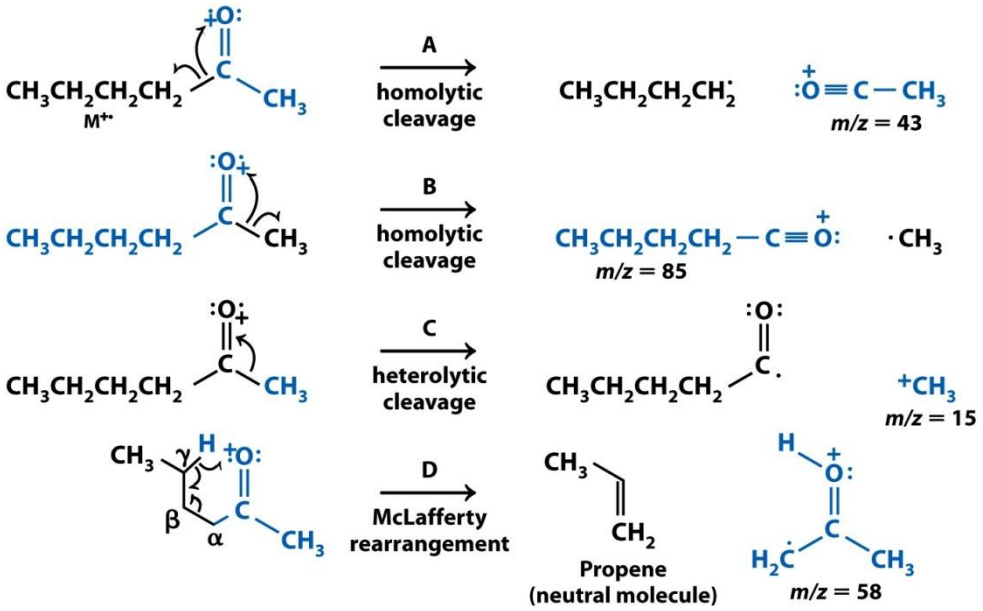
$$\text{No. of C atoms} = \frac{\text{observed } (M+1)/M^{+\cdot} \text{ intensity}}{\text{contribution per carbon atom}} = \frac{6\%}{1.08\%} = 5.6 \approx 6$$

Intensity of M+1 relative to molecular ion for C_nH_m:
Intensity = n x 1.08% + m x 0.012%

Interpretation of mass spectrum

Interpreting fragmentation patterns

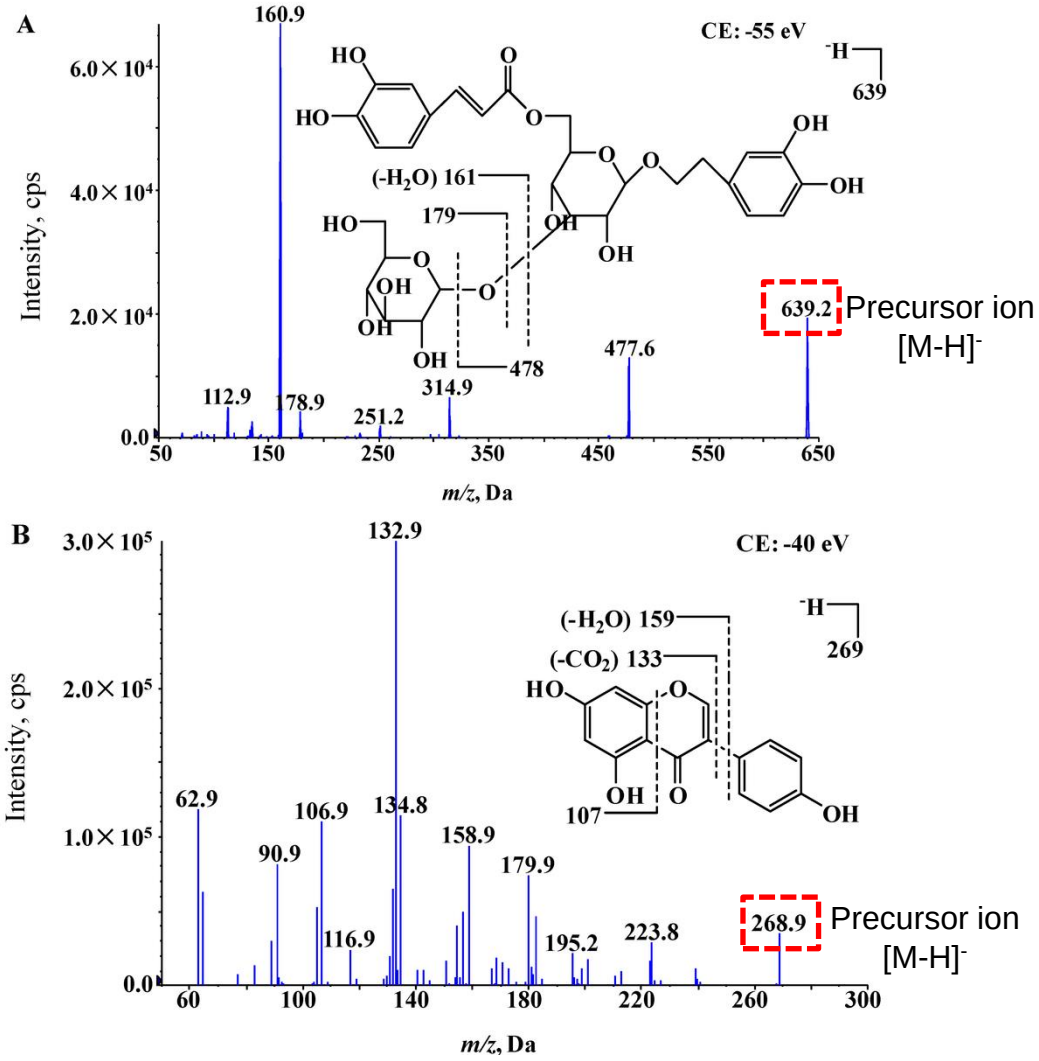




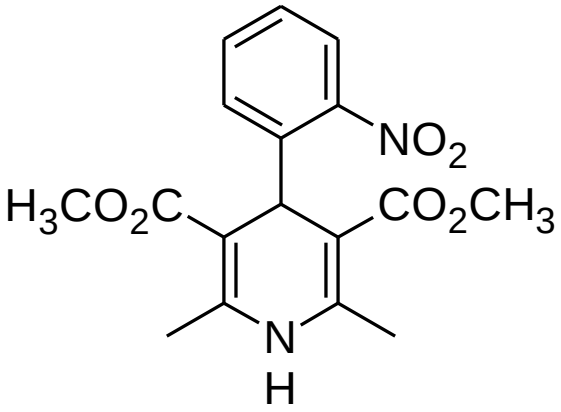
- **Nitrogen rule for (odd-electron) radical ions**
  - ✓ even no. of nitrogen ⇒ even nominal mass
  - ✓ odd no. of nitrogen ⇒ odd nominal mass
- **Nitrogen rule for (even-electron) non-radical ions**
  - ✓ even no. of nitrogen ⇒ odd nominal mass
  - ✓ odd no. of nitrogen ⇒ even nominal mass

Product ion spectrum

Precursor (Parent) ion $\xrightarrow{\text{Fragmentation}}$ Product (Fragment) ions



Nifedipine



- **Molecular formula = $C_{17}H_{18}N_2O_6$**
- **Molecular mass = 346.3346 Da**
- **Nominal mass = 346 Da**
- **Monoisotopic mass (Exact mass) = 346.116486 Da**
- **Mass of e-: 0.00055 Da**
- **$M^{+\bullet}$ = 346.115938 Da**
- **$M^{-\bullet}$ = 346.117035 Da**
- **$[M+H]^+$ = 347.123763 Da**
- **$[M-H]^-$ = 345.10921 Da**

cf. Accurate mass(측정 정밀 질량): 고분해능 질량 분석기(High-Resolution Mass Spectrometer, HRMS)를 사용하여 **실험적으로 측정**된 이온의 질량 값

Isotope pattern ($[M+H]^+$)

m/z	%
347	100
348	19.6
349	3.0
350	0.3

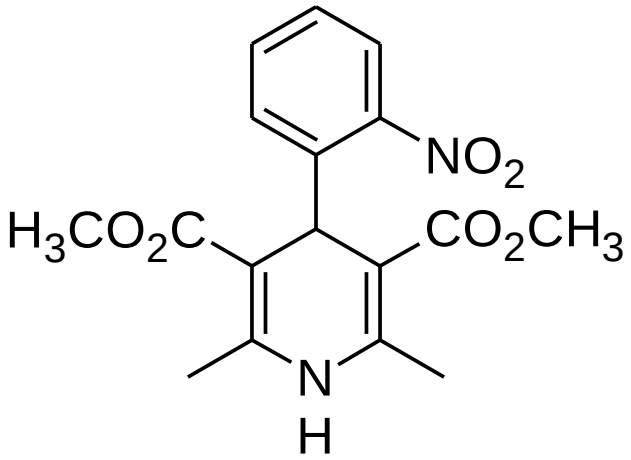
The nitrogen rule:
2 nitrogen ⇒ odd nominal mass

Double bond equivalent

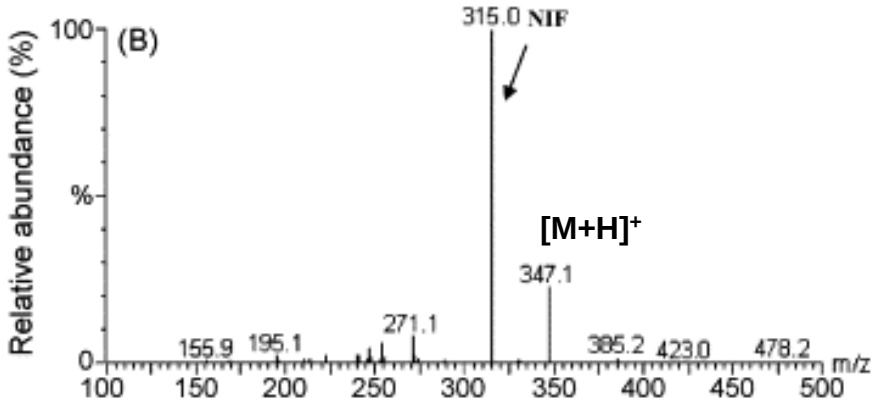
R + DB = c-h/2 + n/2 +1

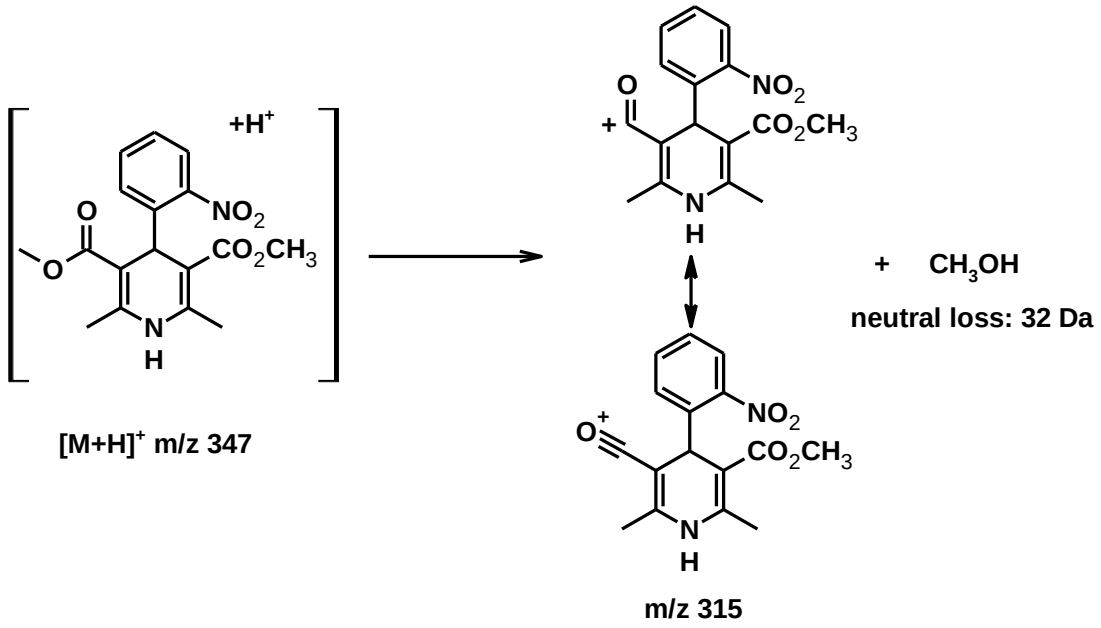
DBE = 17-18/2+2/2+1 = 10

Nifedipine

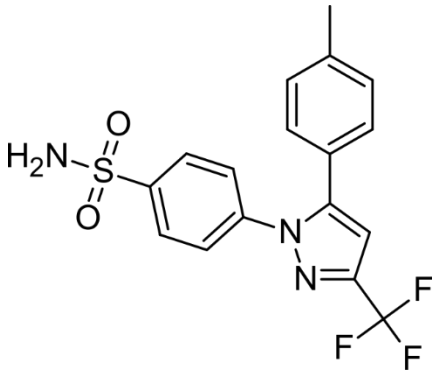


+ESI product ion spectrum

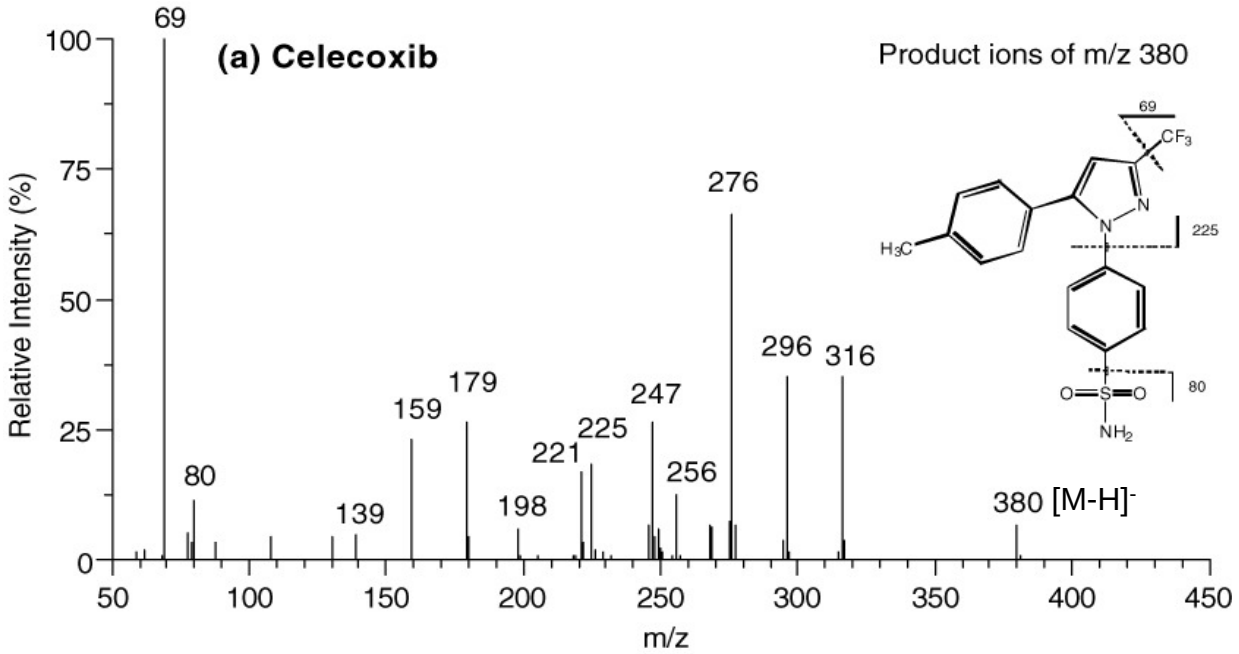




Celecoxib



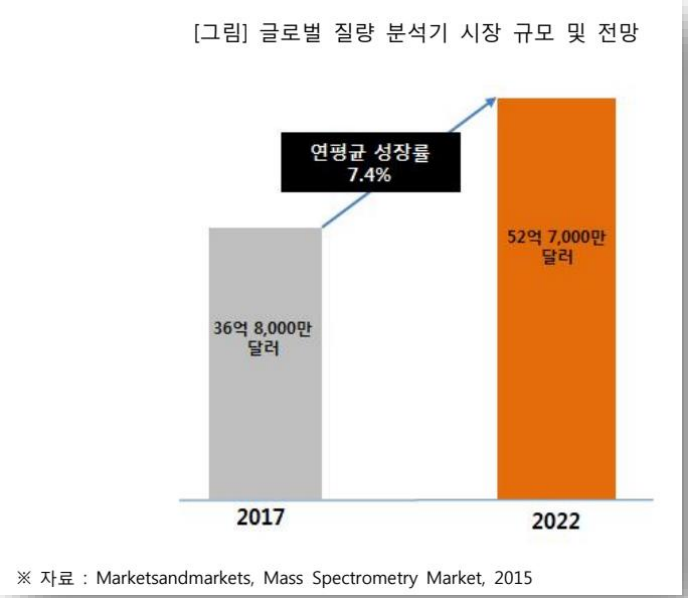
- **Molecular formula = $C_{17}H_{14}F_3N_3O_2S$**
- **Molecular mass = 381.3732 Da**
- **Nominal mass = 381 Da**
- **Monoisotopic mass = 381.0759 Da**
- **Mass of e^- = 0.00055 Da**
- **$M^{+\bullet}$ = 381.0753 Da**
- **$M^{-\bullet}$ = 381.0764 Da**
- **$[M+H]^+$ = 382.0832 Da**
- **$[M-H]^-$ = 380.0686 Da**



What can we do with mass spectra?

- Determination of molecular mass: Molecular/Quasi-molecular ion m/z
- Suggestion of molecular formula: accurate mass and isotope pattern
- Structure characterization: product (fragment) ion spectra
- Protein sequencing
- Metabolomics
- Proteomics
- Quantitation of analytes

……



The Mass Spectrometry Market, 2019-2024

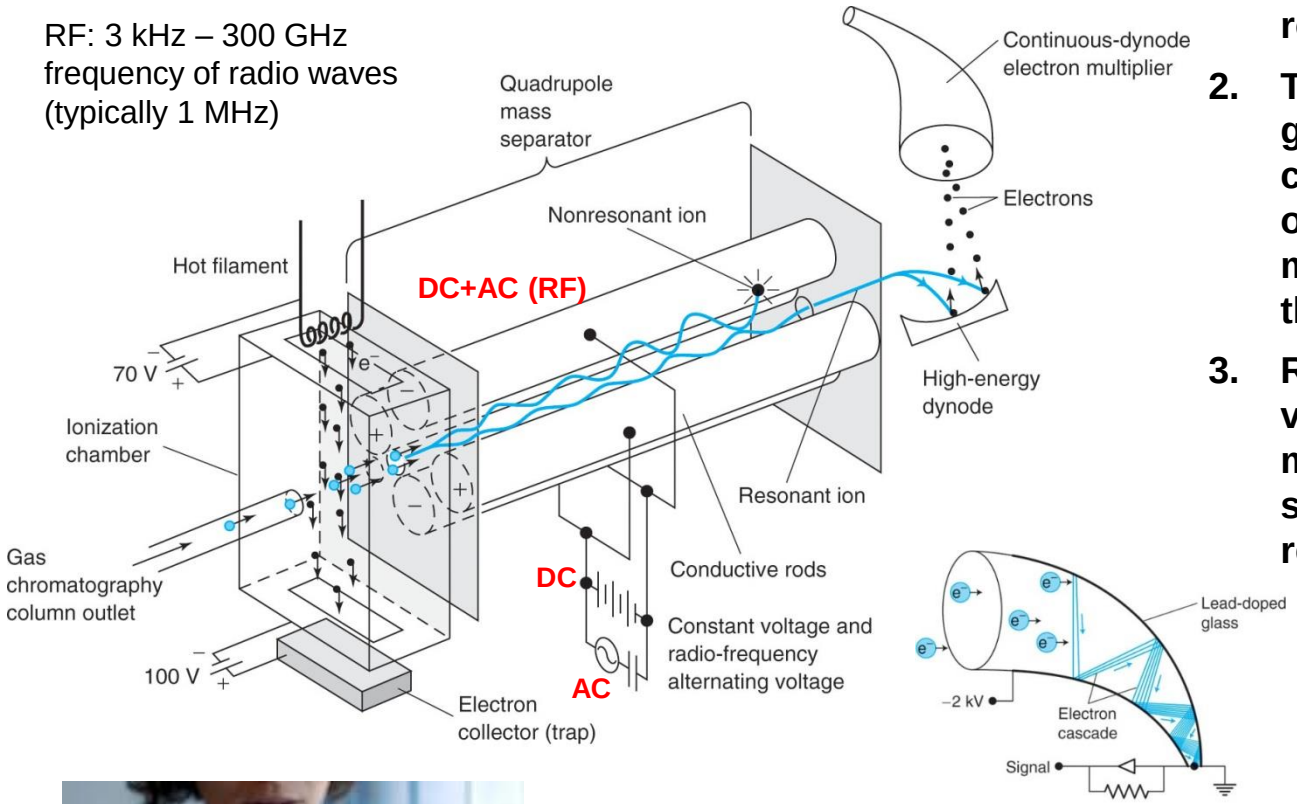
"질량 분석(MS)의 동정/정량 능력은 본질적으로 매우 높고, 과학기기 세계에서는 타의 추종을 불허합니다. 질량 분석 시장은 2019-2024년간 6.1%의 성장률을 보일 것으로 예측되고 있습니다. 시장에서는 Sciex, Thermo Fisher Scientific, Agilent, Waters가 지배적인 지위에 있습니다. 기술 구분별로는 Quadrupole LC/MS가 시스템 전체 매출의 약 24%를 차지하고 있습니다. 이 기술은 제약, 생명공학, 임상시험, 학술연구 등 생명과학 분야의 다양한 용도에서 폭넓게 이용되고 있습니다."

Types of mass spectrometers

Transmission quadrupole mass spectrometer
사중극자 질량분석기

Four parallel rod electrodes
- two positive electrodes facing each other (U+V cos ωt)
- two negative electrodes facing each other (-U-V cos ωt)

RF: 3 kHz – 300 GHz
frequency of radio waves
(typically 1 MHz)

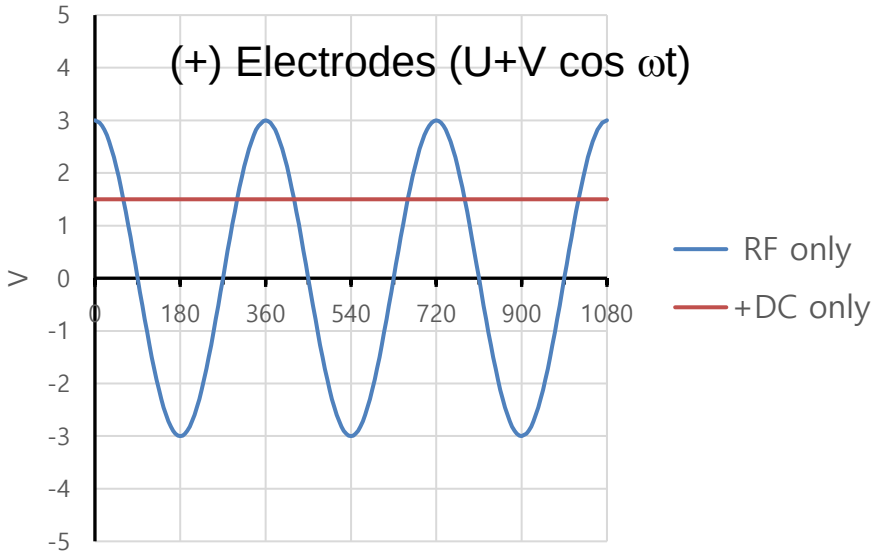


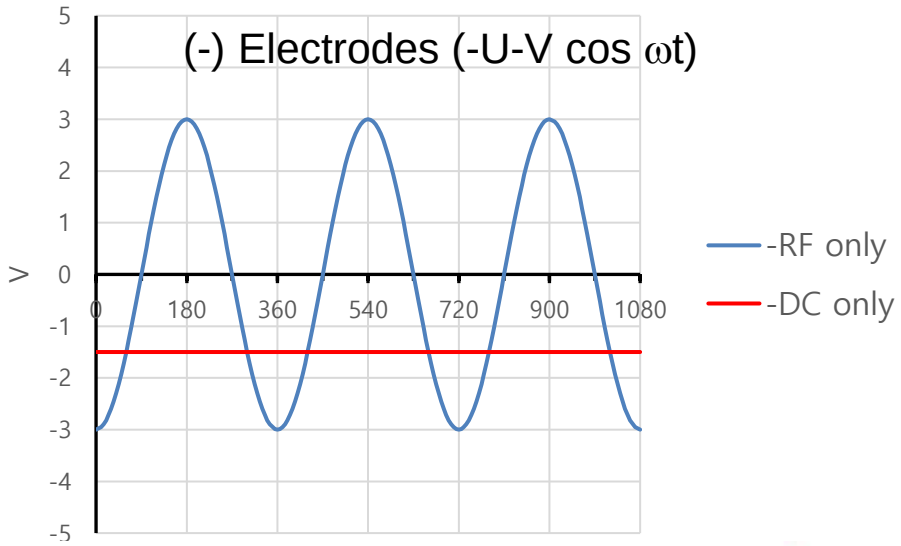
1. **Both a constant voltage (DC) and RF oscillating voltage (AC) are applied to the four parallel rod electrodes.**
2. **The oscillating electric field generated deflects ions in complex trajectories, allowing only ions with one particular m/z ratio (resonant ion) to reach the detector.**
3. **Rapidly varying DC and RF voltages *simultaneously*, maintaining DC/RF constant, select ions of different m/z to reach the detector.**

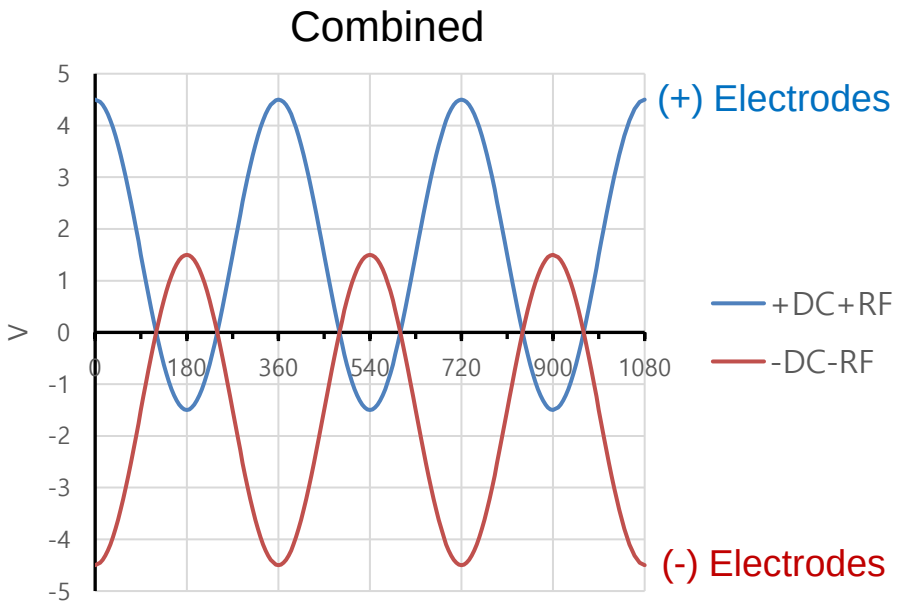


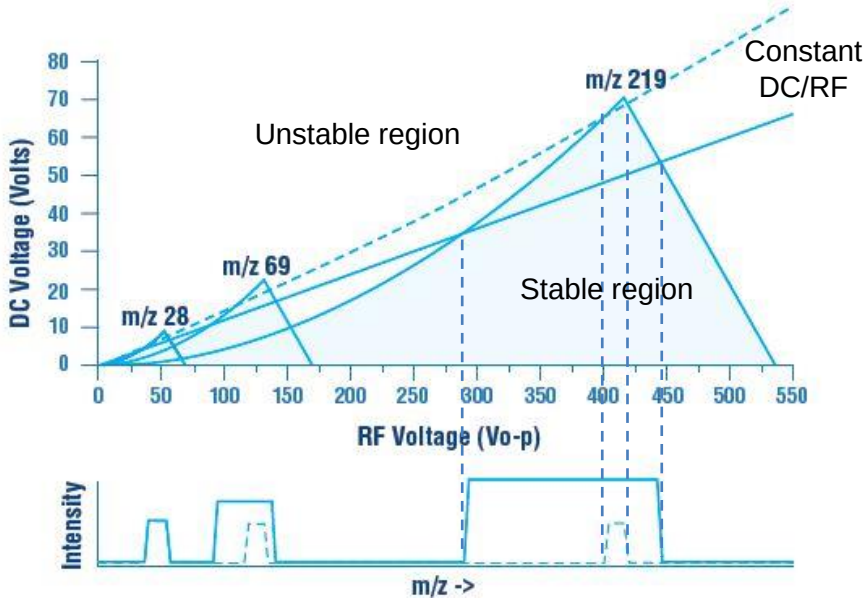
- **2-8 spectra/s covering a mass range: *m/z* 4,000**
- **Resolution: m/z 0.3 (constant resolution), unit resolution**
- **Resolving power varies with *m/z*.**

Transmission quadrupole mass spectrometer





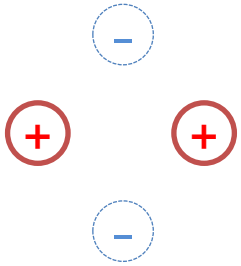


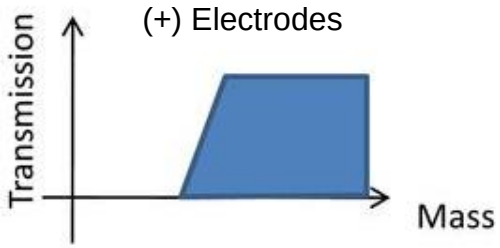


Transmission quadrupole mass spectrometer

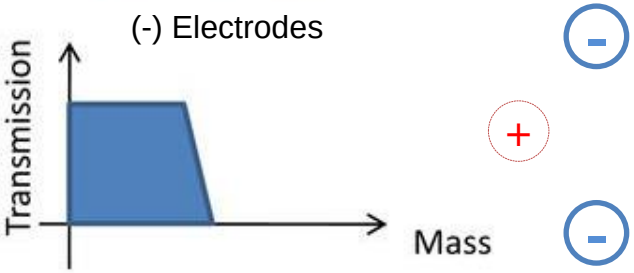
For cations

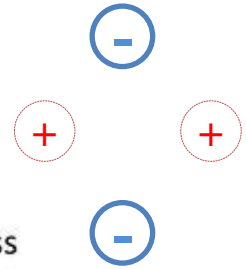
High-mass filter
(특정 m/z 이상 이온 모두 통과)





Low-mass filter
(특정 m/z 이하 이온 모두 통과)





The quadrupole is a band-pass mass filter.



	(+)	(-)
Cations	High-mass filter	Low-mass filter
Anions	Low-mass filter	High-mass filter

➢Properties of the passing band:

- The **width** of the band-pass region (**mass resolution**) is governed by the **V/U ratio**.
- The **mass** at the centre of the band-pass region depends on the **magnitude** of both AC (V) and DC (U) **potentials**.

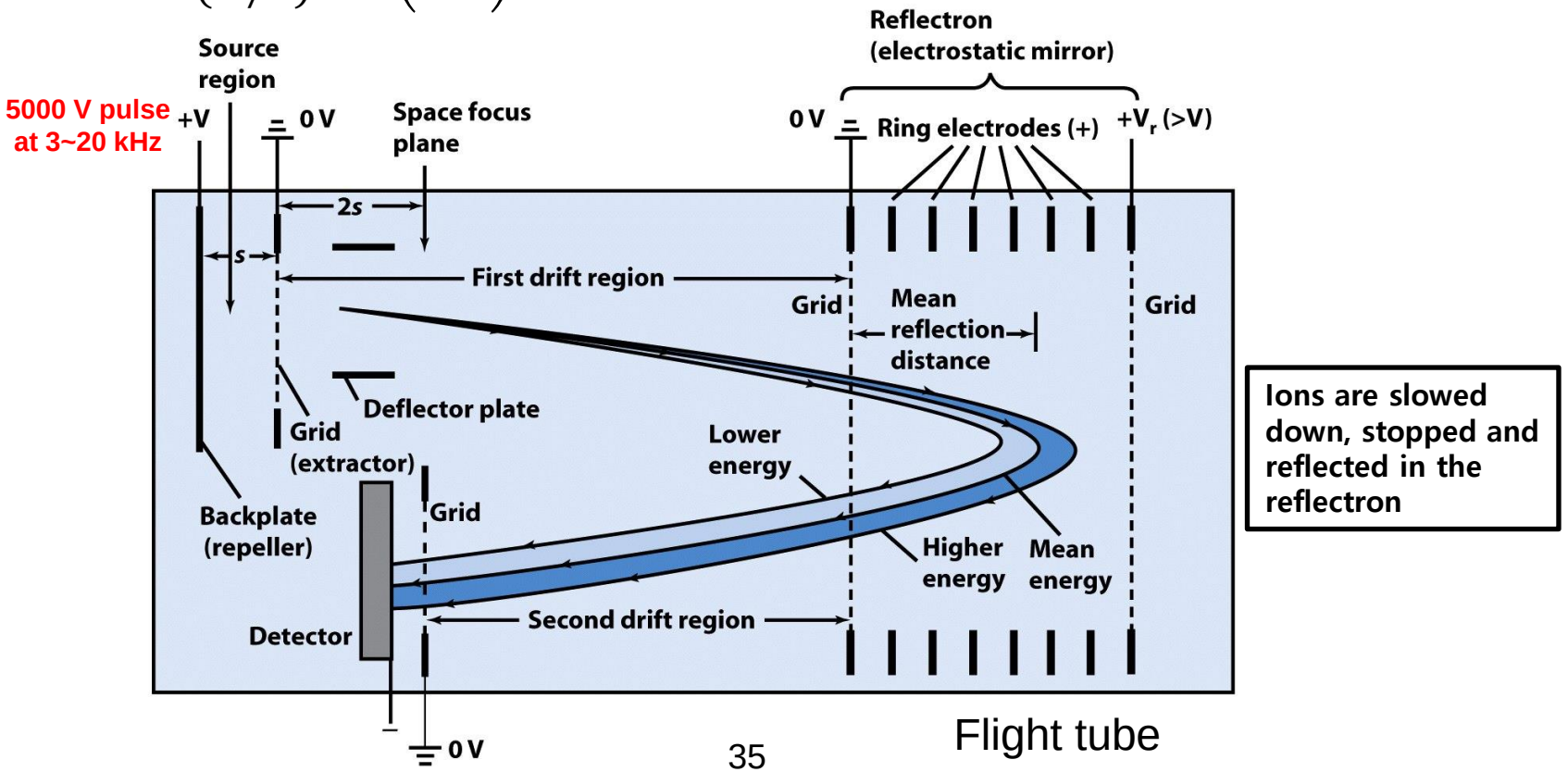
Types of mass spectrometers

Time-of-flight (TOF) mass spectrometer
비행시간형 질량분석기

$$E_k = \frac{1}{2}mv^2 = q(C)\cdot E(V) = zeV \;\Rightarrow\; v = \sqrt{\frac{2E_k}{m}} = \sqrt{\frac{2zeV}{m}}$$

$$\frac{m}{z} = \frac{2eV}{v^2} = \frac{2eV}{(L/t)^2} = \left(\frac{2eV}{L^2}\right)\times t^2 = A\times t^2$$

E_k: kinetic energy
m: mass of ion
v: velocity of ion
ze: charge of ion where e is the magnitude of the charge of an electron
V: electric potential difference



Ions are slowed down, stopped and reflected in the reflectron

Flight tube

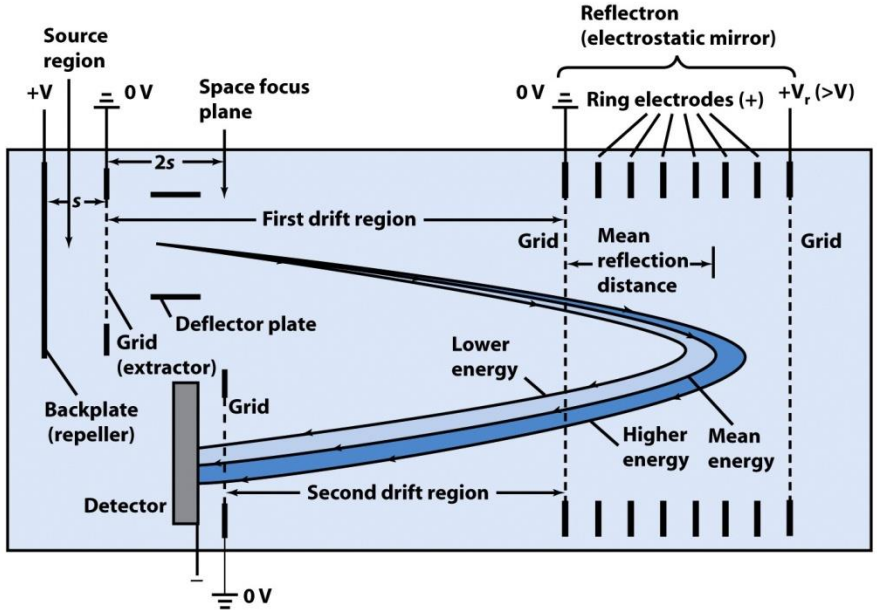
Types of mass spectrometers

Time-of-flight (TOF) mass spectrometer

1. The ion near the backplate gains more kinetic energy ⇒ Ions emerge from the source with varying kinetic energy (i.e., kinetic energy spread) ⇒ Heavier ions with more-than-average kinetic reach the detector at the same time as lighter ions with less-than-average kinetic energy ⇒ low resolving power
2. All ions of a given mass reach the space focus plane at the same time. After the space focus plane, ions diverge again, with faster ions overtaking slower ions.
3. The reflectron is a series of ring electrodes held at increasingly positive potential, terminated by a grid whose with a potential more positive than that of the backplate. Ions entering the reflectron are slowed down, stopped, and reflected back to the left.
4. The more kinetic energy an ion had , the further it penetrates before it is turned around
5. Reflected ions reach a new space focus plane at the grid in front of the detector: all ions of the same mass reach the detector at the same time, regardless of their initial kinetic energies ⇒ high resolution.



- High-resolution mass spectrometer (HRMS) (Resolving power 25,000, mass accuracy ~0.001 Da)
- Mass range: ~m/z 10^6
- High acquisition rate: 10^2-10^4 spectra/s



Types of mass spectrometers

Operating pressure requirements for various types of mass spectrometer

High resolution, high vacuum!

Mass spectrometer	Operating pressure		
	Bar	*Torr*	*Pa*
Transmission quadrupole	**~10^{-9}**	**~10^{-6}**	**~10^{-4}**
Time-of-flight	**~10^{-12}**	**~10^{-9}**	**~10^{-7}**
3D QIT	**~10^{-6}**	**~10^{-3}**	**~10^{-1}**
Orbitrap	**~10^{-13}**	**~10^{-10}**	**~10^{-8}**

1 atm : standard sea level pressure
1 atm = 760 torr = 760 mmHg
1 bar = 750 torr = 10^5 Pa = 100 kPa = 1000 hPa
1 atm = 760/750 bar = 1.01325 bar = 1013.25 mbar
1 bar = 14.5038 psi (pound-force per square inch)

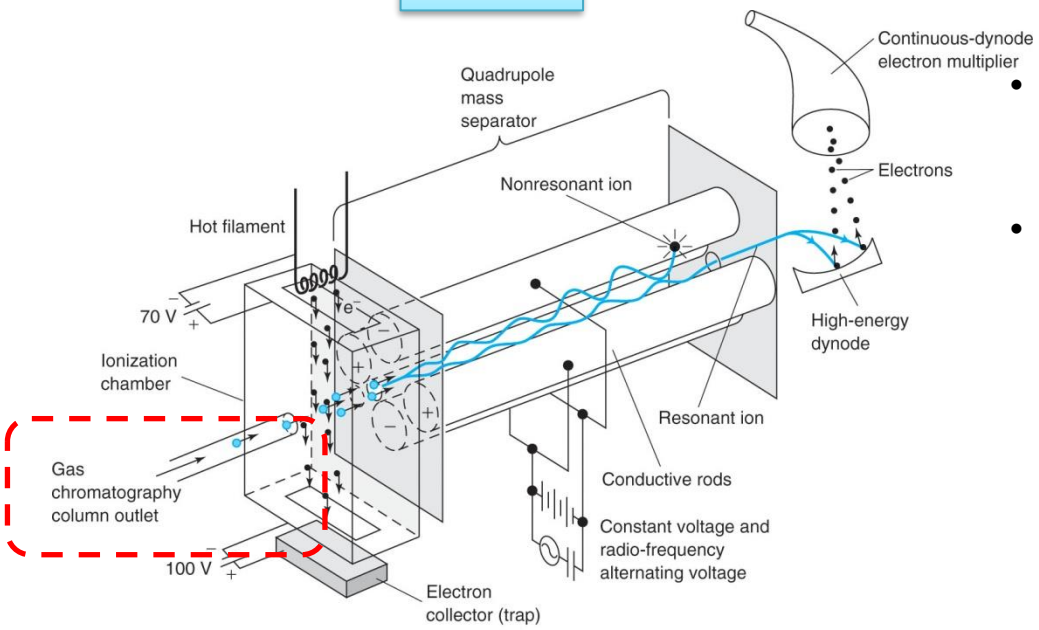
Chromatography-Mass spectrometry

Gas chromatography or Liquid chromatography | Ion source | Mass analyzer | Detector

High pressure **High vacuum**

Hyphenated technique

GC-MS



Capillary column is connected directly to the MS inlet

LC-MS

- **LC generates a huge volume of gas at the interface between LC and MS, that must be removed prior to ion introduction to MS**
- **Nonvolatile mobile-phase additives must be avoided**

API

(Atmospheric Pressure Ionization)

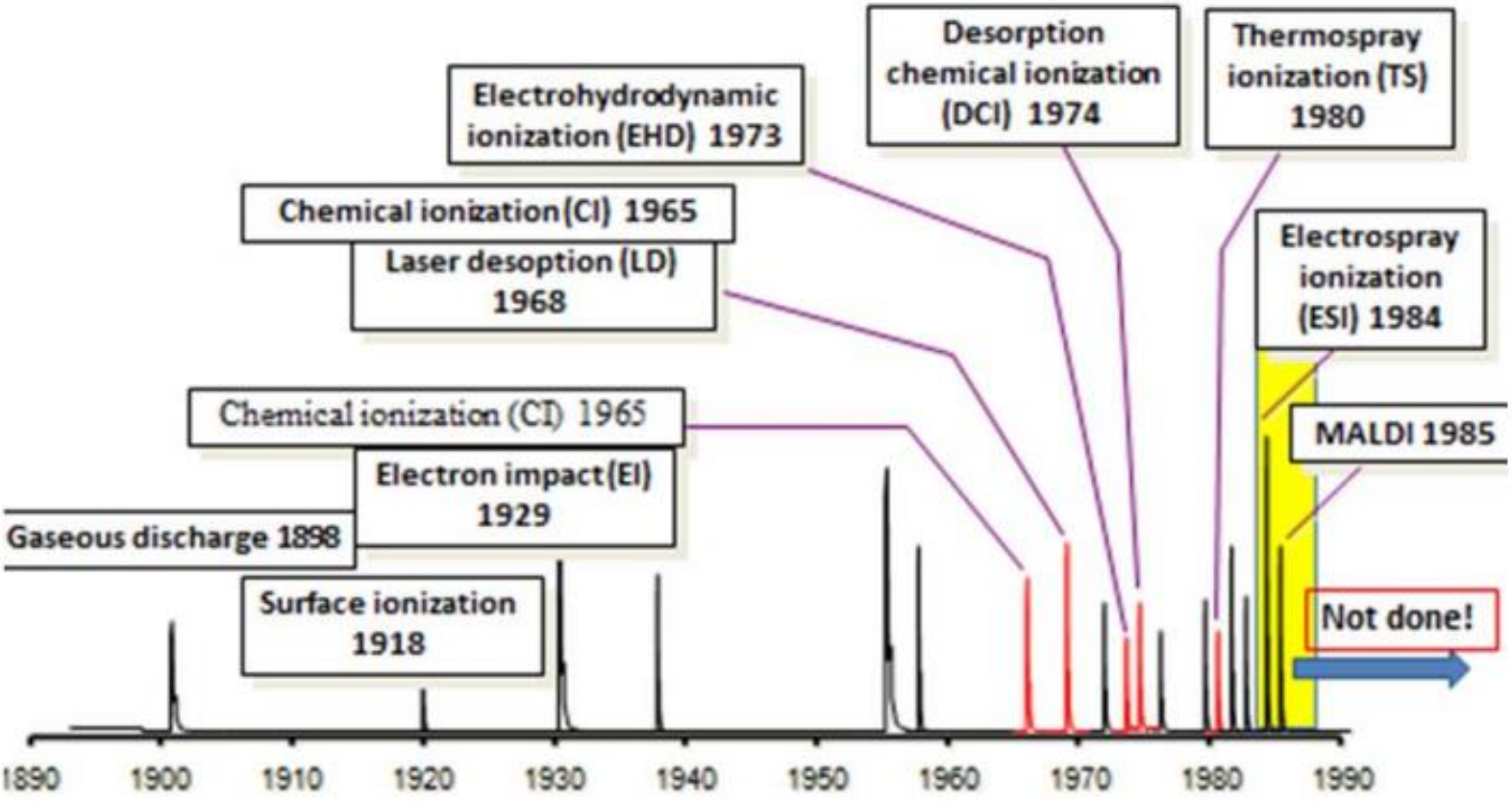
ESI

(Electrospray Ionization)

APCI

(Atmospheric Pressure Chemical Ionization

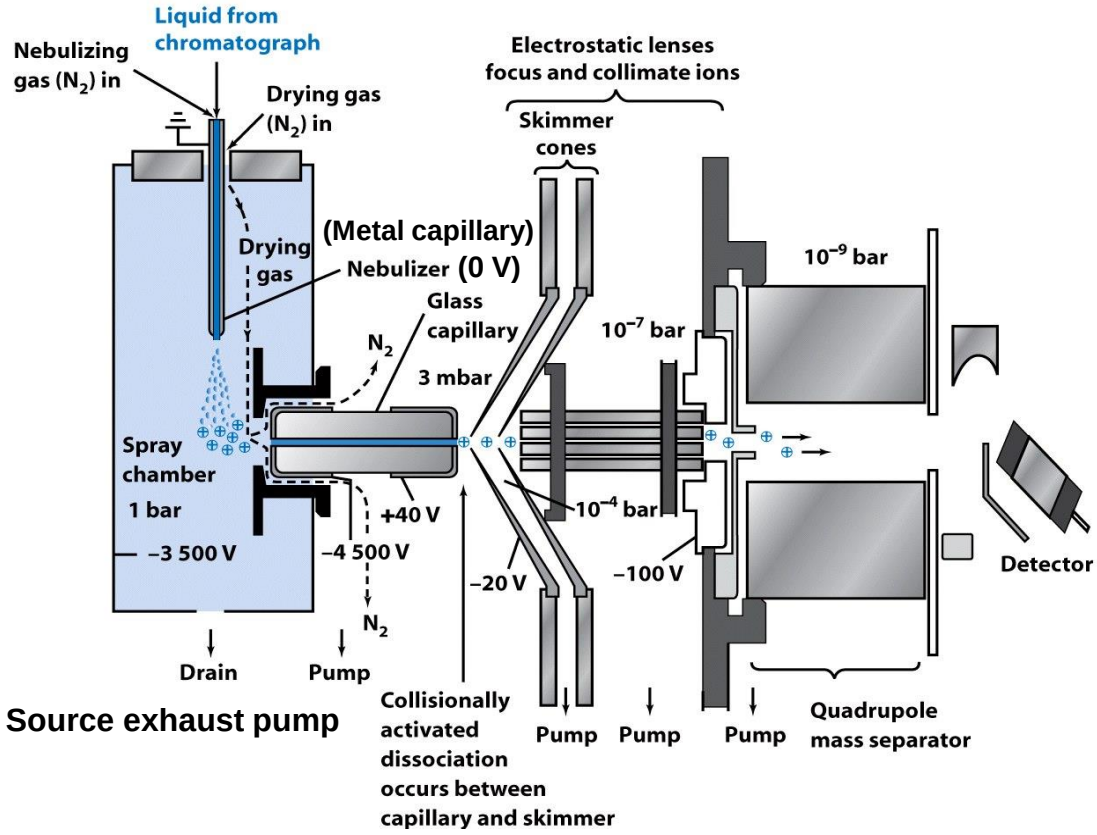
Ionization Methods : Time Line



- Gas chromatography (GC): EI, CI
- Liquid chromatography (LC): API (ESI, APCI, APPI)

Chromatography-Mass spectrometry

ESI (ElectroSpray Ionization, 전기분무 이온화)



- Liquid from chromatography column enters the steel nebulizer along with a coaxial flow of N_2 gas
- The nebulizer is held at 0 V and the spray chamber is held at -3.5 kV (for positive ion MS)
- The strong electric field at the nebulizer tip, combined with the coaxial flow of N_2 gas, creates a fine aerosol of charged particles
- Ions that vaporize from aerosol droplets were already in solution in the chromatography column
- Positive ions from the aerosol are attracted toward the glass capillary leading into the mass spectrometer by a more negative potential (-4.5 kV)
- Gas flowing in the spray chamber transports ions to the right through the glass capillary to the exit

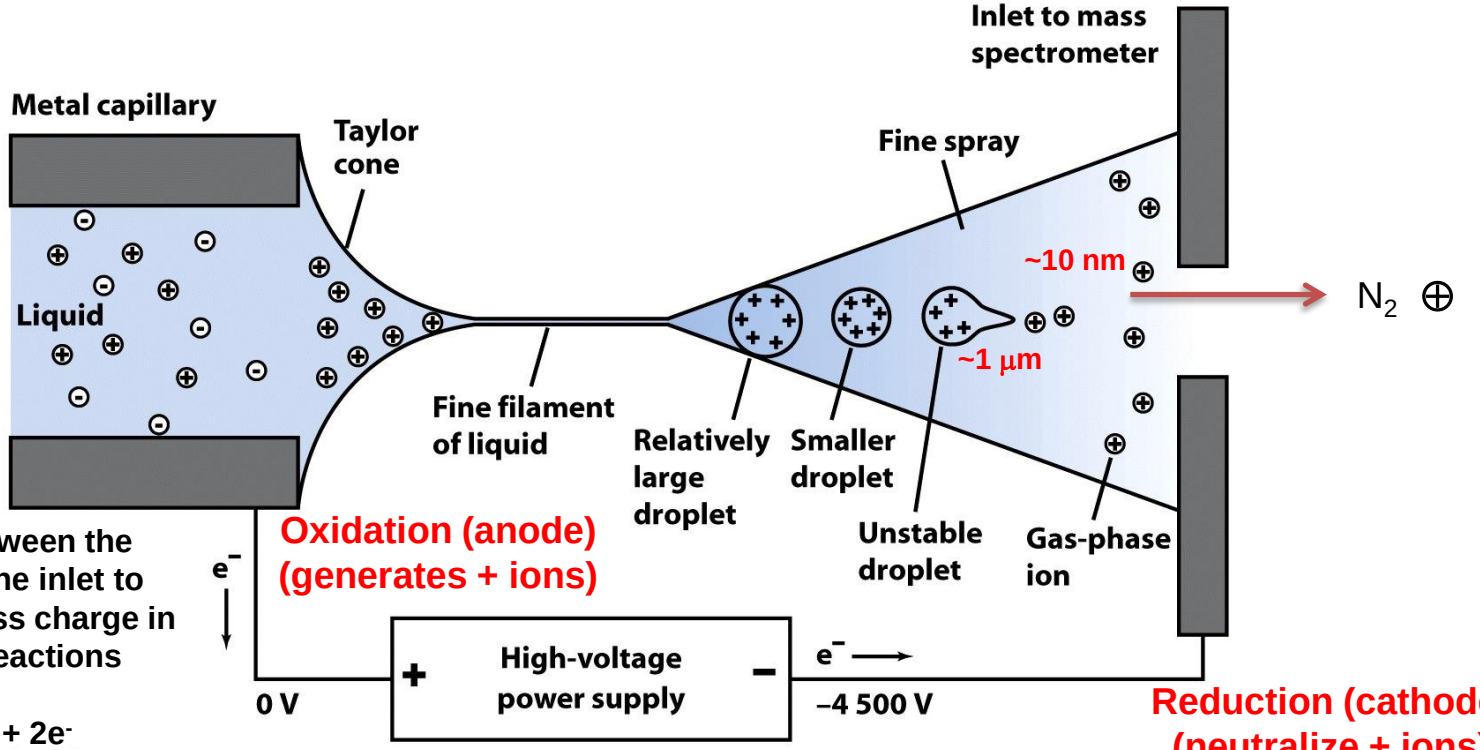
John B. Fenn, shared Nobel prize in Chemistry (2002)

Chromatography-Mass spectrometry

ESI (Electrospray Ionization)

Adduct ions

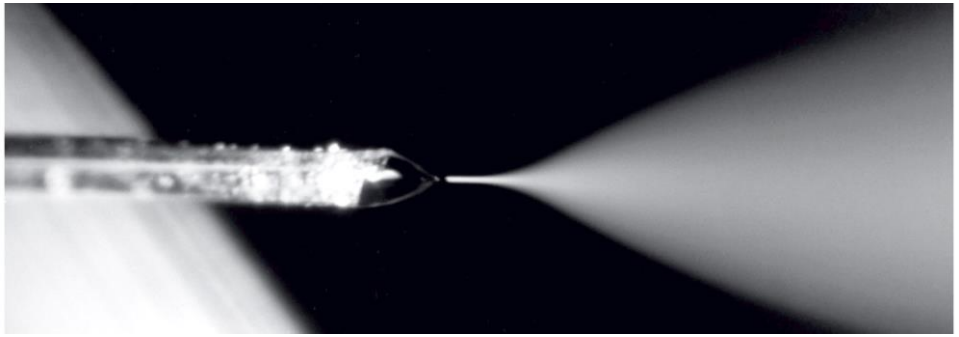
- MH^+ (m/z=M+1)
- $M(NH_4^+)$ (m/z=M+18)
- $M(Na^+)$ (m/z=M+23)
- $M(K^+)$ (m/z=M+39)
- $[M-H]^-$ (m/z=M-1)
- $M(HCO_2^-)$ (m/z=M+45)
- $M(CH_3CO_2^-)$ (m/z=M+59)
- .....

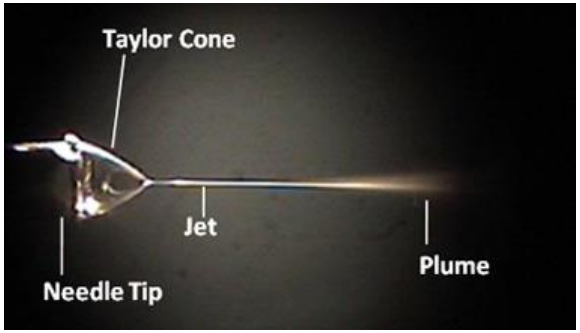


Voltage imposed between the metal capillary and the inlet to the MS creates excess charge in the liquid by redox reactions

$Fe(s) \rightarrow Fe^{2+} + 2e^-$

$H_2O(l) \rightarrow 1/2O_2 + 2H^+ + 2e^-$

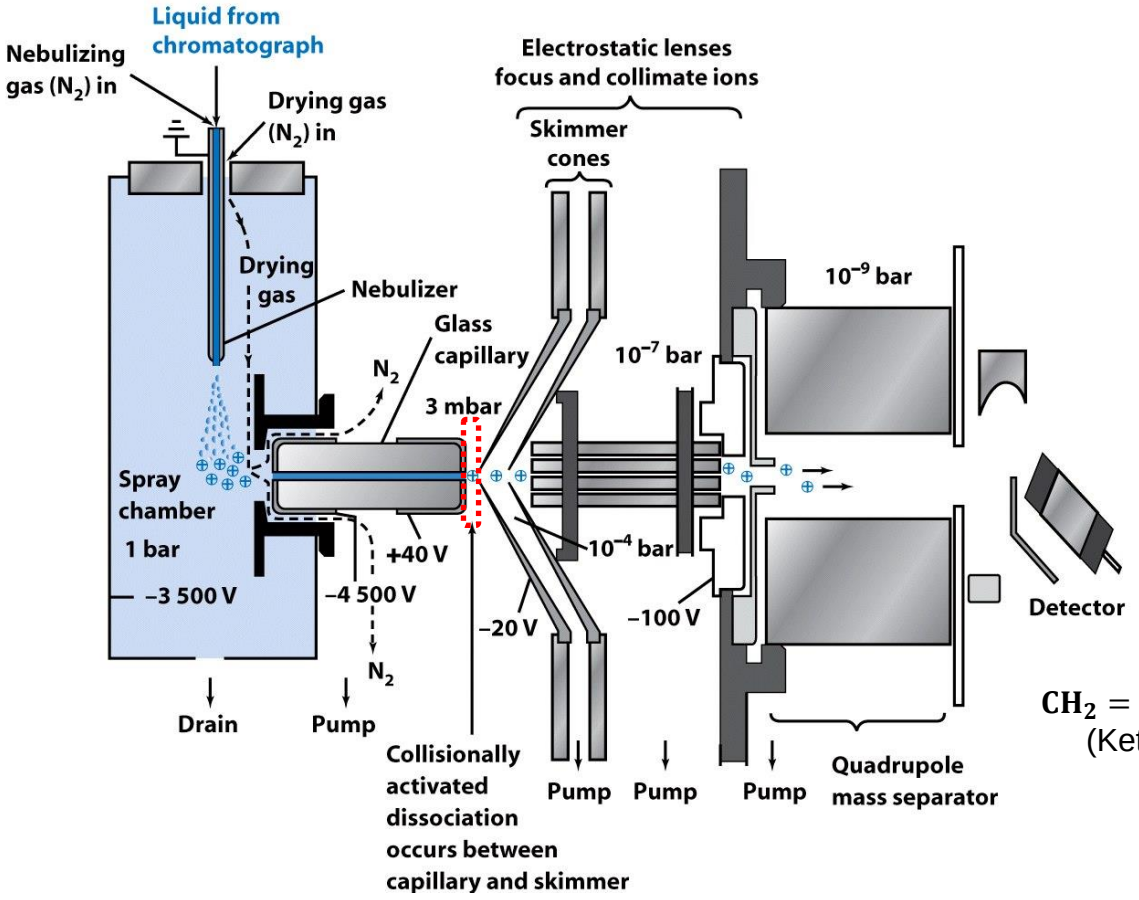




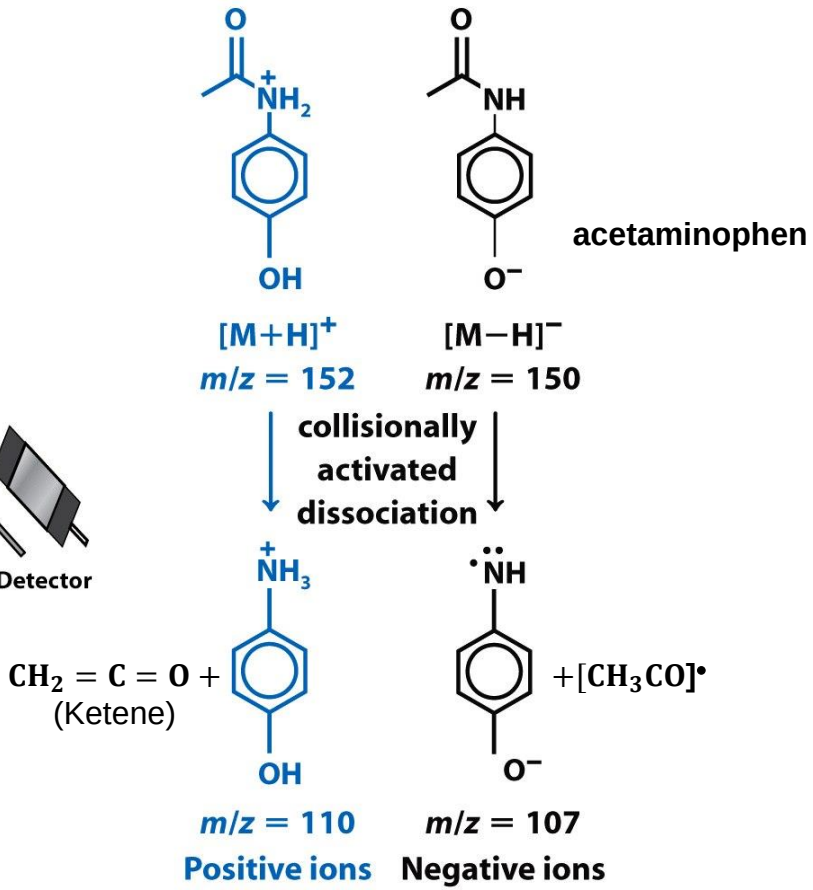
Chromatography-Mass spectrometry

ESI (Electrospray Ionization)

Collisionally activated dissociation (CAD) or Collision-induced dissociation (CID)

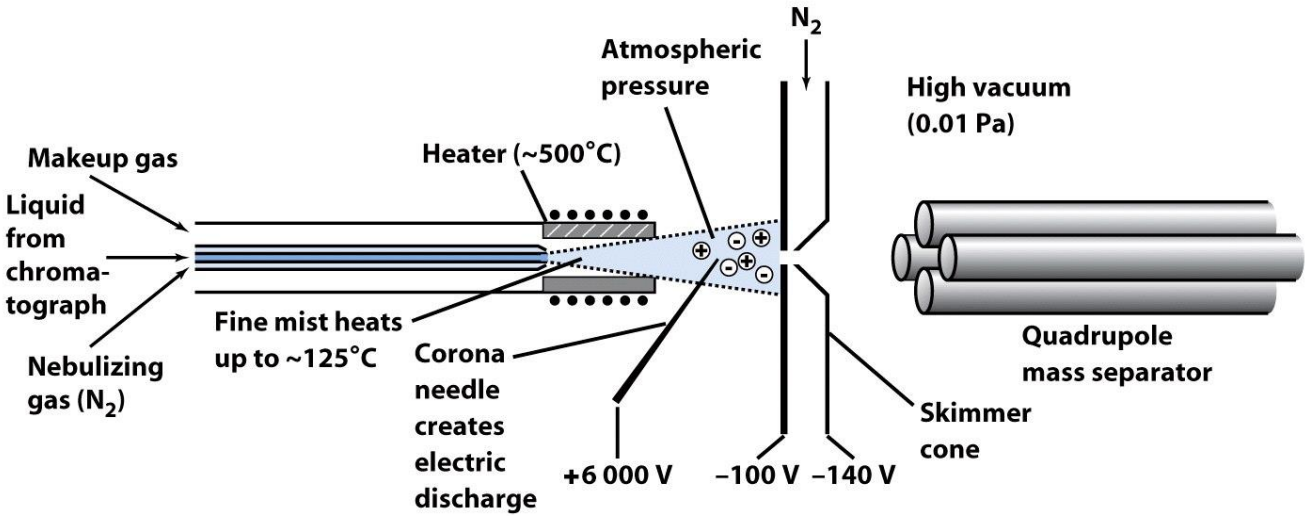


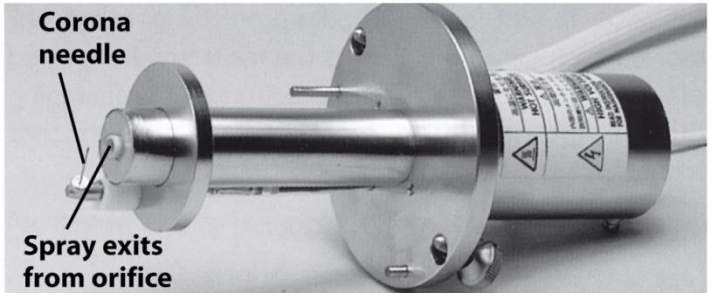
Collision between analyte ions and N_2 gas



Chromatography-Mass spectrometry

APCI (Atmospheric Pressure Chemical Ionization)





- Chromatography eluate is converted into a fine mist by heat and a coaxial N_2 flow
- Solvent and analyte evaporate from the mist
- The electric discharge from the high-voltage corona needle creates gaseous ions from the analyte

$$N_2 + e^- \rightarrow N_2^{+\cdot} + 2e^-$$

$$N_2^{+\cdot} + 2N_2 \rightarrow N_4^{+\cdot} + N_2$$

$$N_4^{+\cdot} + H_2O \rightarrow H_2O^{+\cdot} + 2N_2$$

$$H_2O^{+\cdot} + H_2O \rightarrow H_3O^+ + {}^{\cdot}OH$$

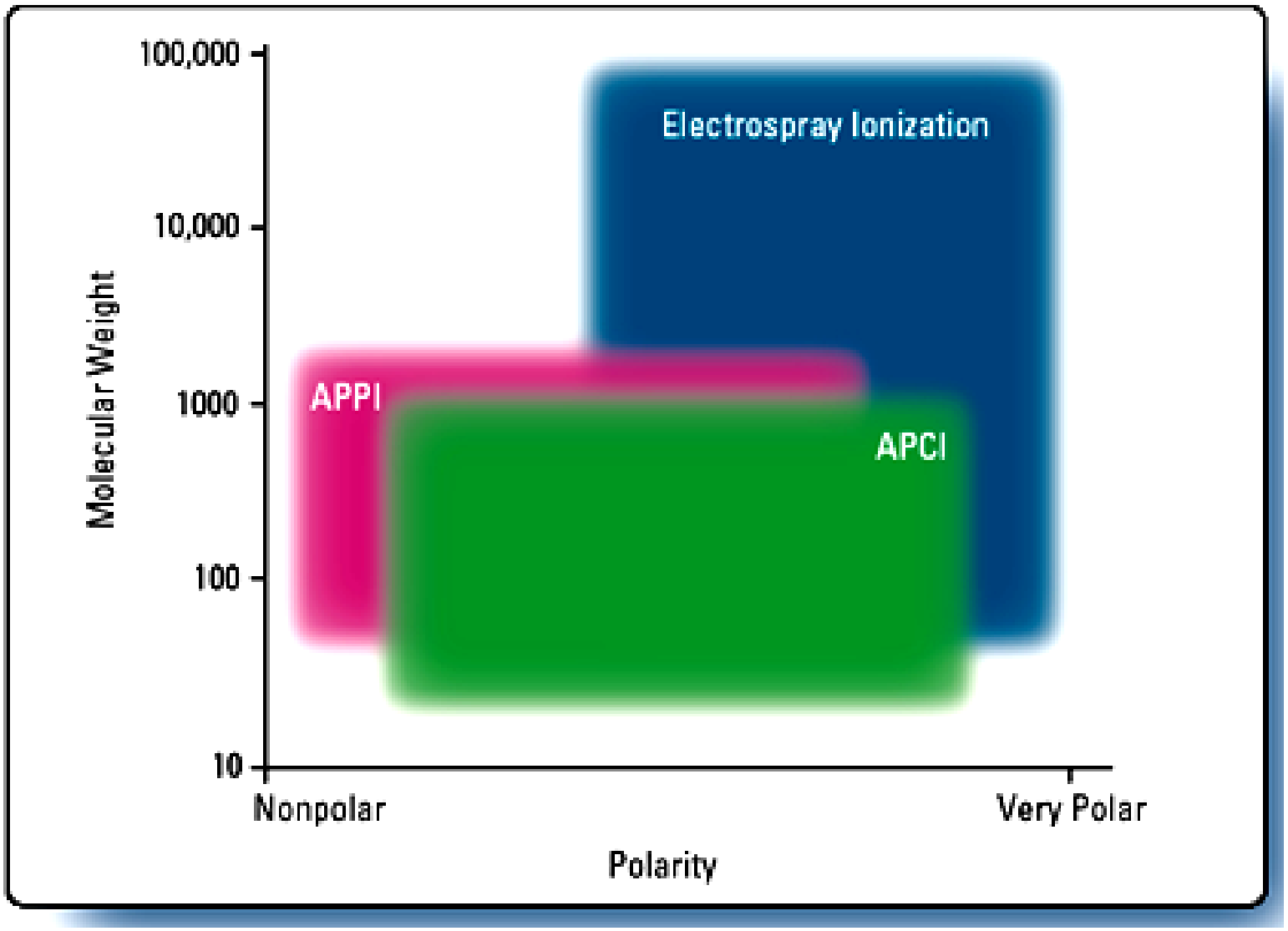
$$H_3O^+ + nH_2O \rightarrow H_3O^+(H_2O)_n$$

$$H_3O^+(H_2O)_n + M \rightarrow MH^+ + (n+1)H_2O$$

Chromatography-Mass spectrometry

Comparison between ESI and APCI

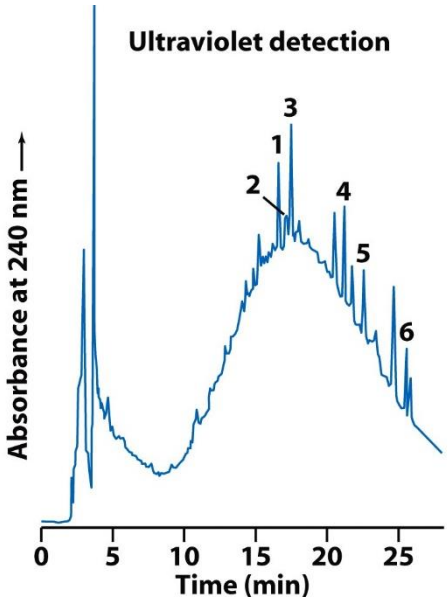
	ESI	APCI
Ionization phase	Solution	Gas
Acceptable flow rate	Up to 0.4 mL/min	Up to 2 mL/min
CAD capability	Yes	Yes
Single-charge ion formation	Yes	Yes
Multiple-charge ion formation	Yes	No
Suitability for macromolecules	Yes	No
Analyte volatility	No	Yes


100,000
10,000
1000
100
10
Molecular Weight
Electrospray Ionization
APPI
APCI
Nonpolar
Very Polar
Polarity

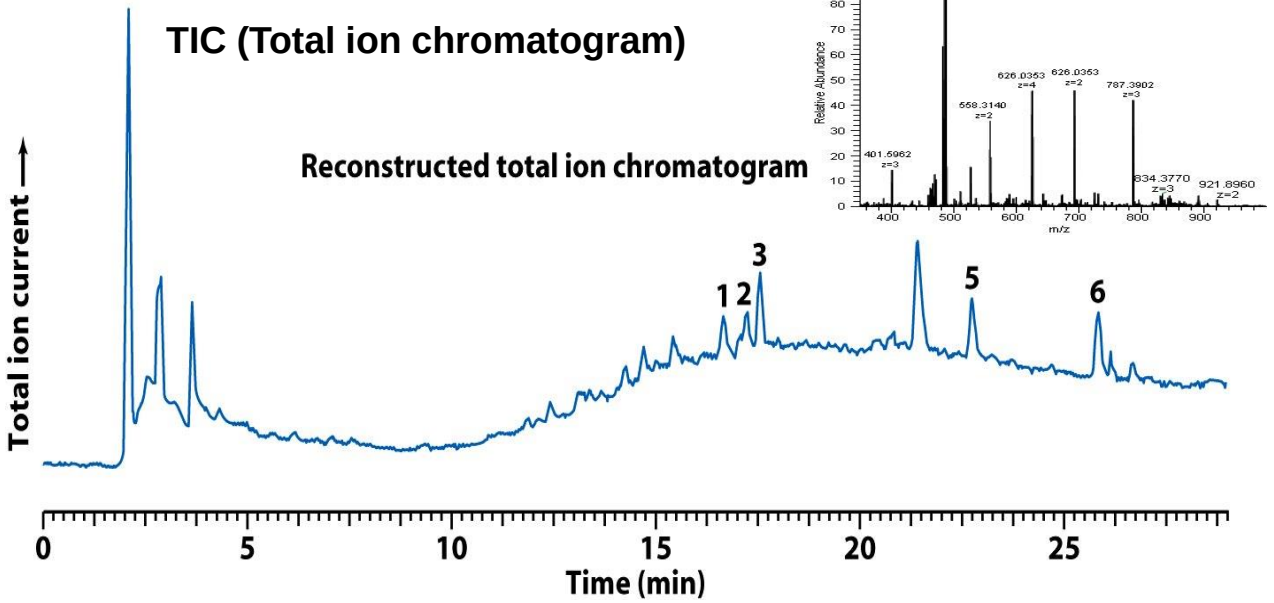
Chromatography-Mass spectrometry

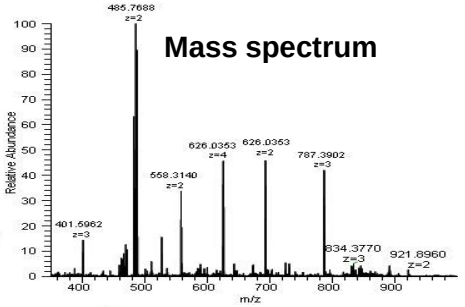
River water spiked with herbicides (1~6) at ~1 ppb

LC-UV



LC-MS

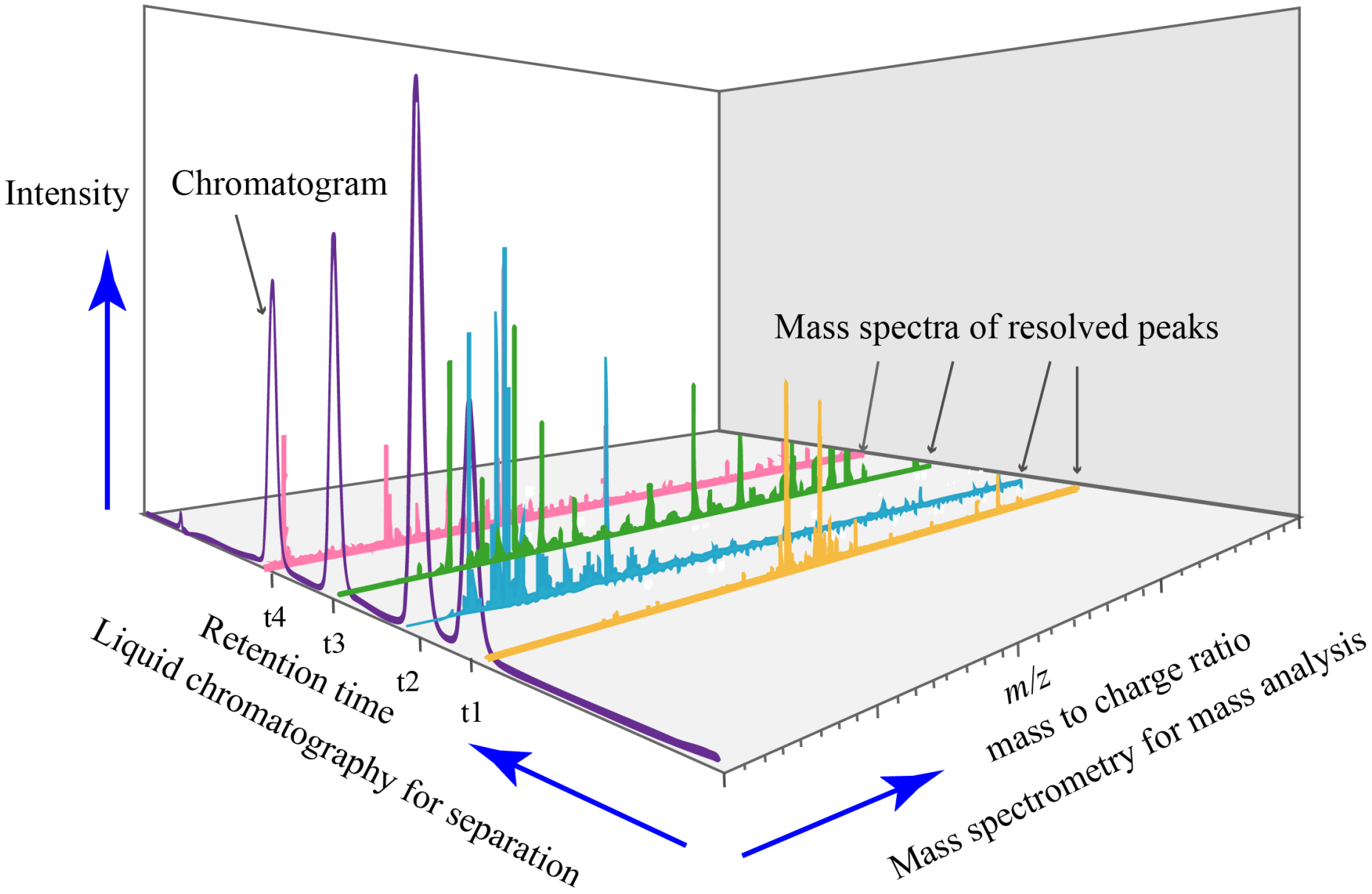




Time (min)	Abs @240 nm
1	3
2	25
3	15
.....	.....
30	51

Time (min)	Intensity (Ion current)									
	m/z 50	51	52	53	54	55	...	499	500	Total
1	1	3	2	4	5	1	4	10	2	32
2	2	3	1	5	6	9	3	12	3	44
3										
.....										
30										

Full scan mass spectra


Intensity
Chromatogram
Mass spectra of resolved peaks
t4
t3
t2
t1
Retention time
Liquid chromatography for separation
m/z
mass to charge ratio
Mass spectrometry for mass analysis

Chromatography-Mass spectrometry

Selected ion monitoring (SIM) and extracted ion chromatogram (EIC)

SIM: mass spectrometer is set to monitor just a few values of m/z

Time (min)	Intensity (Ion current) (m/z 312)
1	3
2	5
3	25
.....	.....
30	12

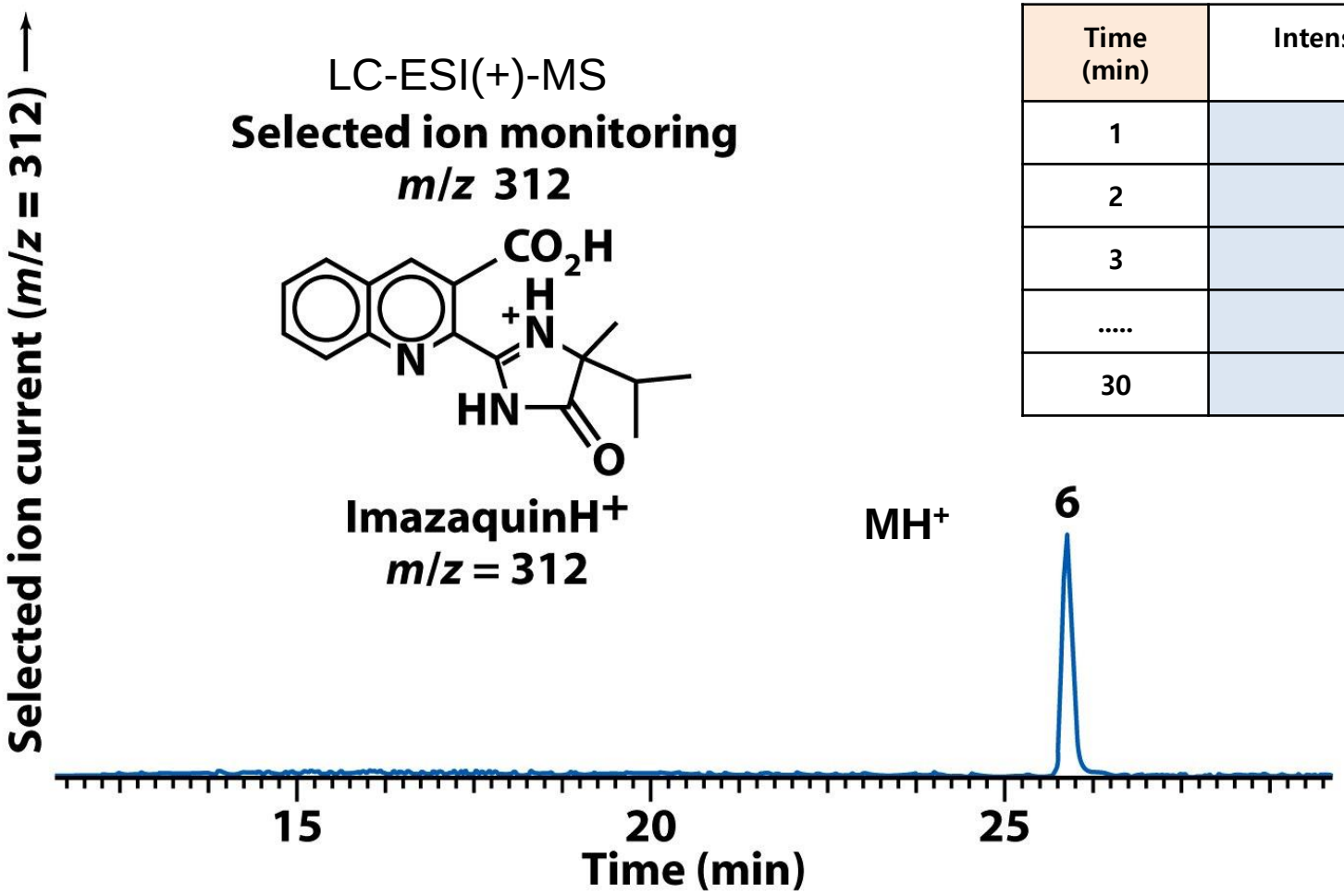
LC-ESI(+)-MS

Selected ion monitoring

***m/z* 312**

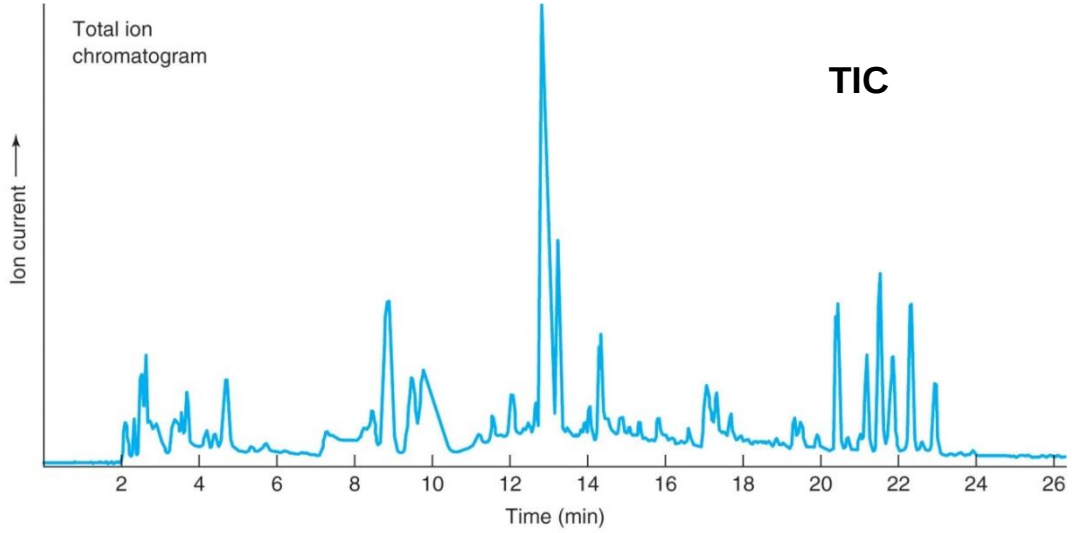
ImazaquinH$^+$

***m/z* = 312**



Chromatography-Mass spectrometry

Selected ion monitoring (SIM) and extracted ion chromatogram (EIC)

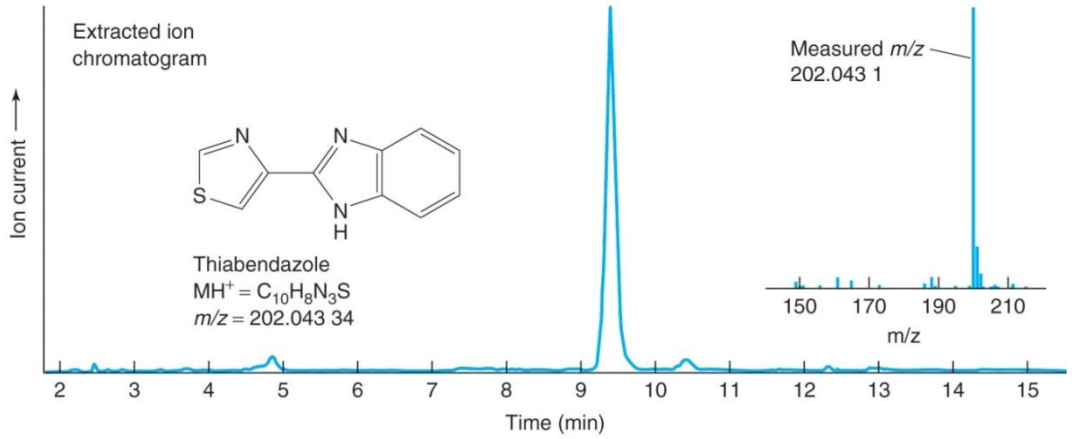


(a)

Time (min)	Intensity (Ion current)					
	50	.....	202	.....	500	Total
1						
2						
3						
.....						
30						

Full scan mass spectra

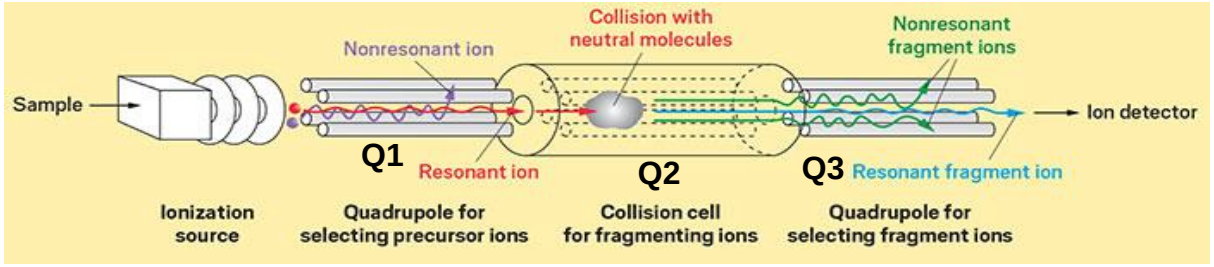
EIC or XIC



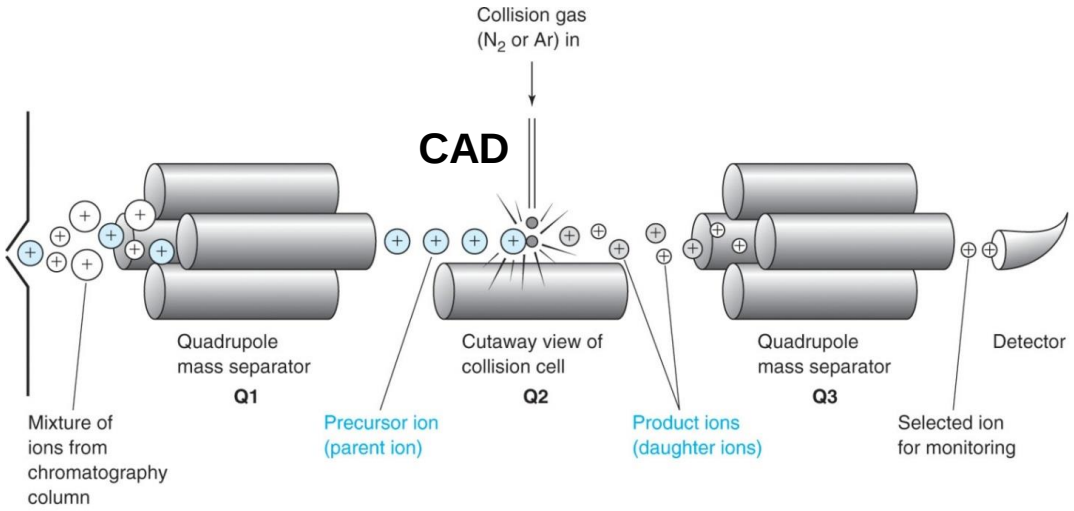
(b)

Chromatography-Mass spectrometry

Tandem mass spectrometry (MS/MS or MS^n)

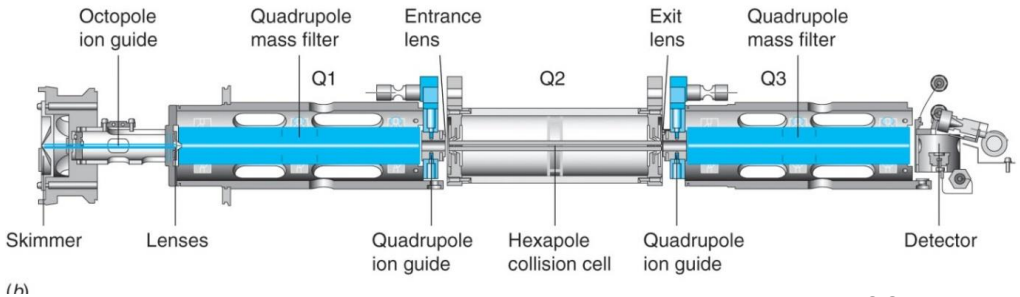


Triple quadrupole mass spectrometer
삼중 사중극자 질량분석기





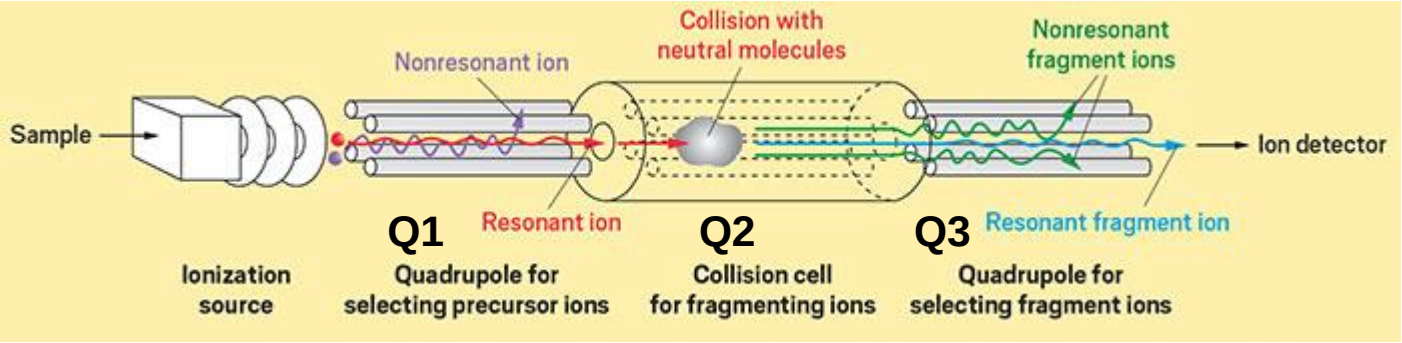
Tandem in space
cf. Tandem in time

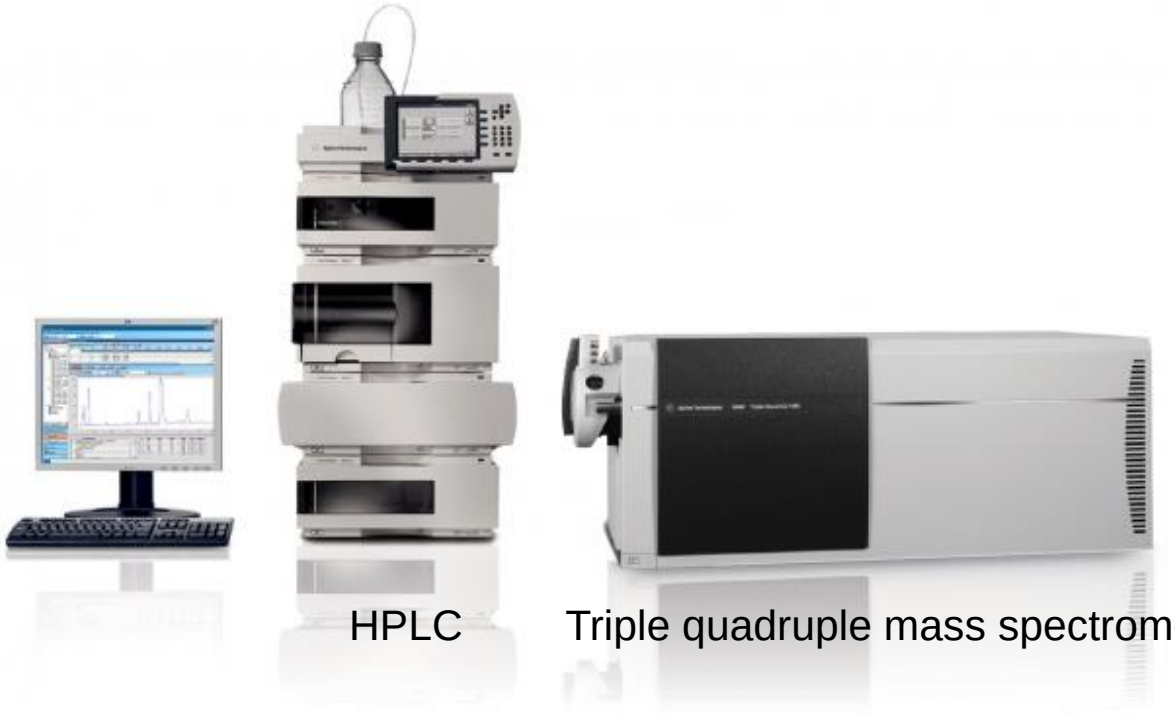


The process in Q2 is called *collisionally activated dissociation*.

CAD (Collision-activated dissociation)
= CID (Collision-induced dissociation)

Triple quadrupole mass spectrometer



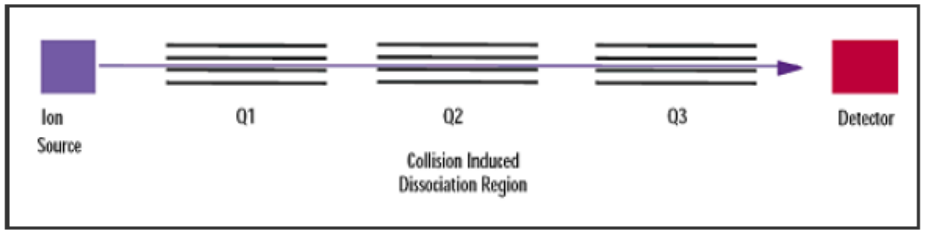


HPLC Triple quadruple mass spectrometer

Agilent 6460 triple quadrupole (QQQ) LC-MS system

Chromatography-Mass spectrometry

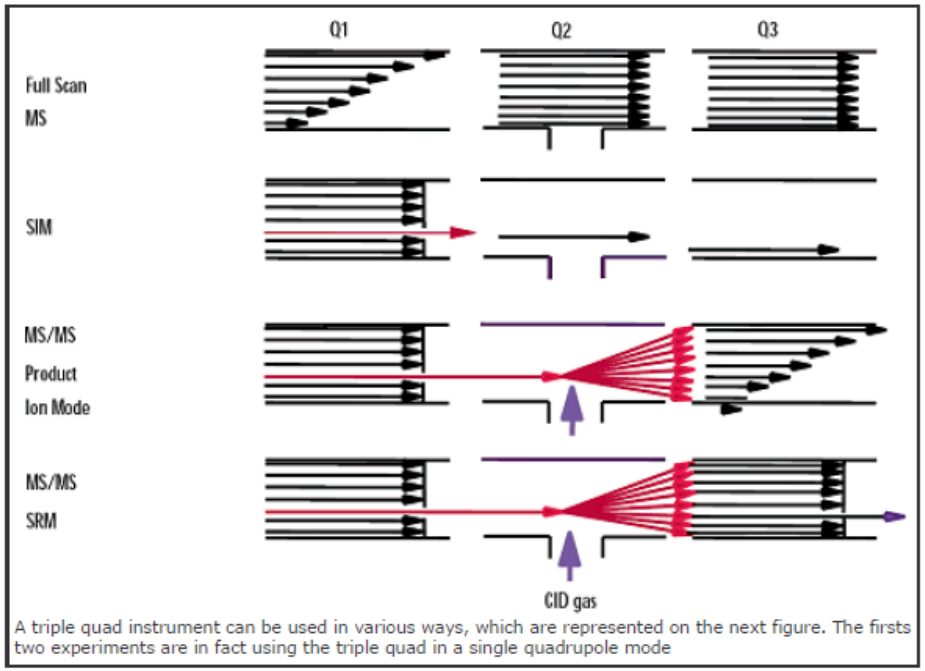
Triple quadrupole scan modes



Q1 & Q3: RF only (no mass filtering), scan or SIM
Q2: Collision gas on or off

⇒ 3(Q1) x 2(Q2) x 3(Q3) = 18 modes?

Back to Top



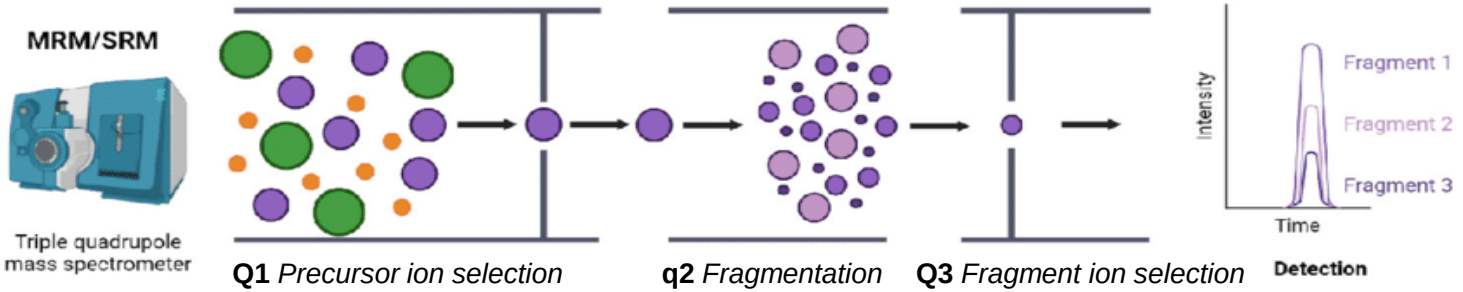
A triple quad instrument can be used in various ways, which are represented on the next figure. The firsts two experiments are in fact using the triple quad in a single quadrupole mode

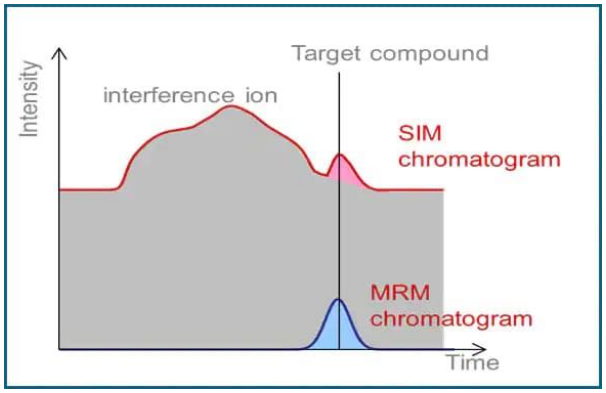
Scan mode	Q1	Q2	Q3
Full scan	Scan	Off	RF only
Selected ion monitoring (SIM)	SIM	Off	RF only
Product ion scan	SIM	On	Scan
Selected reaction monitoring (SRM)	SIM	On	SIM
Precursor ion scan	Scan	On	SIM
Neutral loss scan	Scan	On	Scan

Chromatography-Mass spectrometry

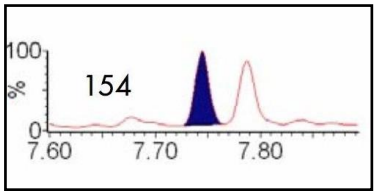
Selected reaction monitoring (SRM) or multiple reaction monitoring (MRM)

(One of the most powerful quantitative analytical methods currently available)

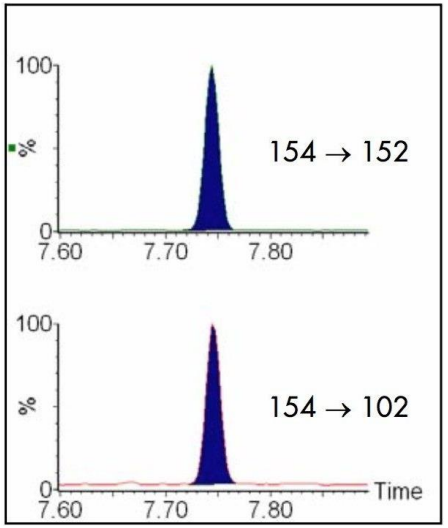


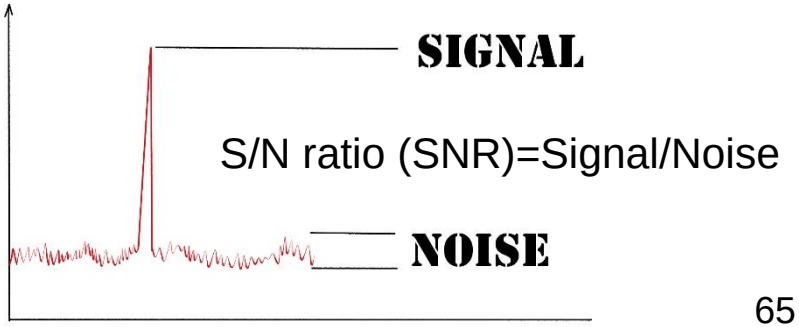


SIM



SRM/MRM





Chromatography-Mass spectrometry

Selected reaction monitoring (SRM) or multiple reaction monitoring (MRM)

(One of the most powerful quantitative analytical methods currently available)

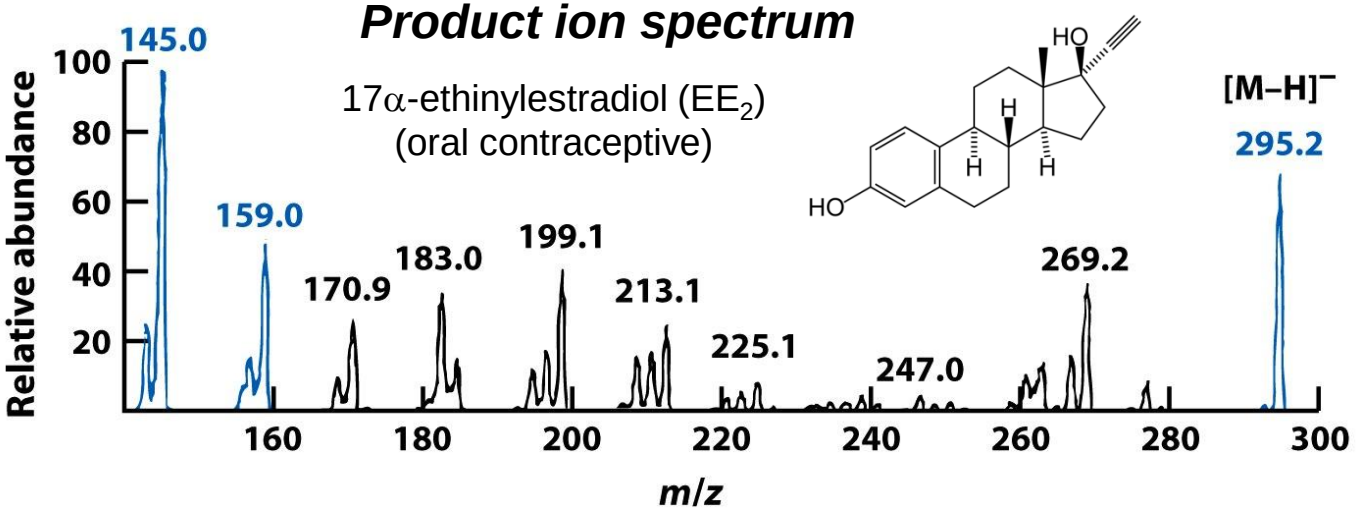
Q1: SIM

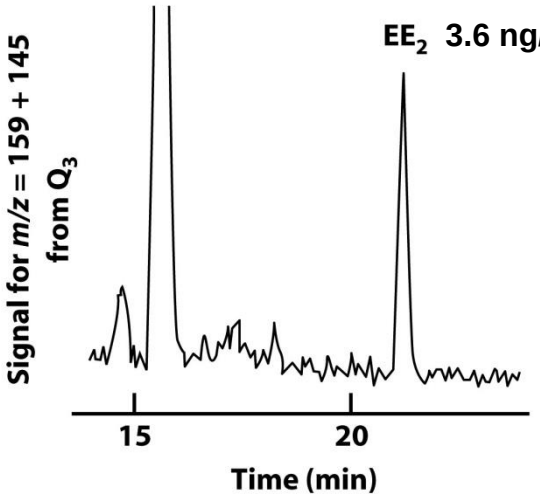
⇓

Q2: Gas on

⇓

Q3: SIM





SRM chromatogram (m/z 295→159+145)

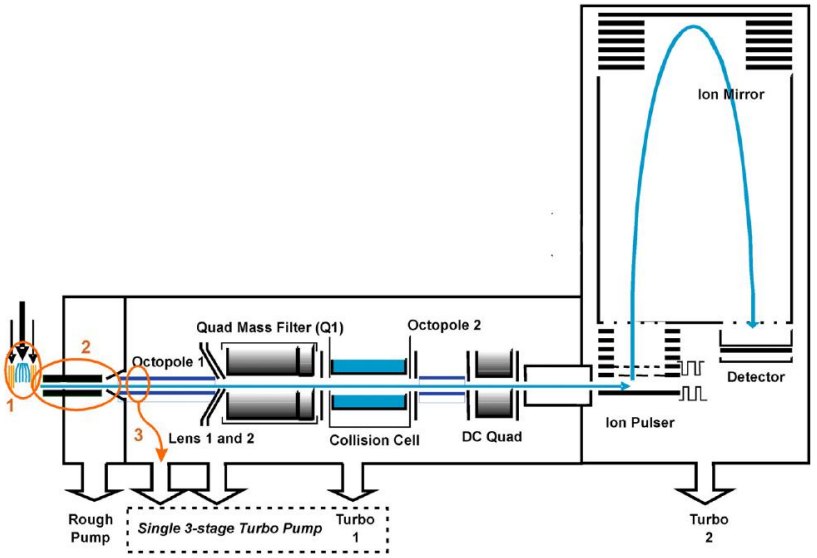
Q1: SIM (m/z 295)
Q3: SIM (m/z 159 & 145)

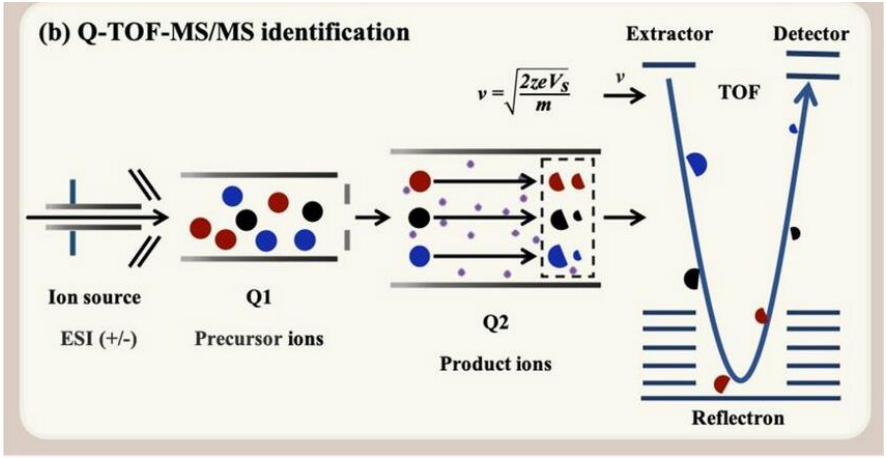
Time	Intensity					
	145	159				Total
1						
2						
3						
.....						
30						

Chromatography-Mass spectrometry

Quadrupole Time-of-Flight (Q-TOF) LC/MS system



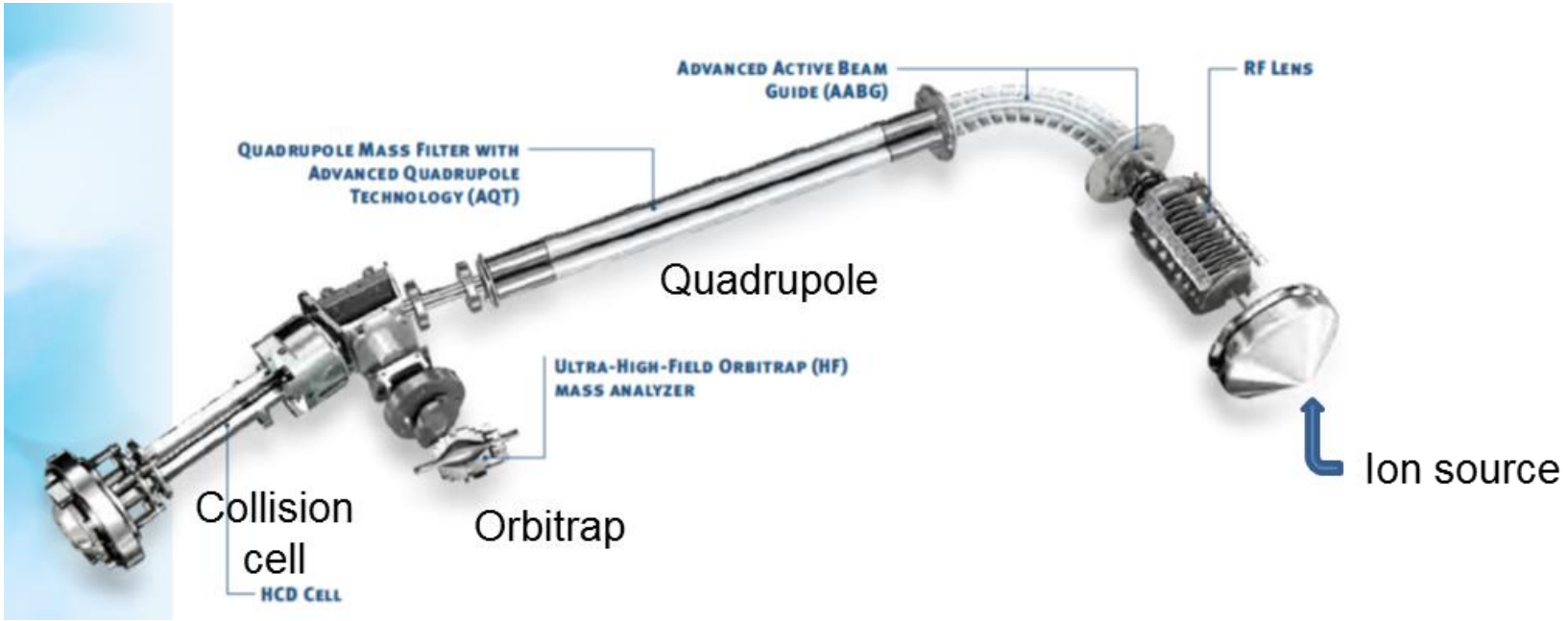




Chromatography-Mass spectrometry

Quadrupole-Orbitrap mass spectrometer

Thermo Scientific™ Q Exactive™ HF hybrid quadrupole-Orbitrap mass spectrometer



Chromatography-Mass spectrometry

Quadrupole-Linear ion trap (Q-Trap) mass spectrometer

